EDUCATIVE COMMENTARY ON ISI 2018 MATHEMATICS PAPERS

(Last Updated on 02/07/2018)

Contents

Introduction

The first draft of the Commentary Paper 1 was uploaded on 20/05/2018. Commentary on Paper 2 is now added to it. Also the solution to Q.2 in Paper 1 is corrected, that to Q.19 revised and that to Q.29 in Paper 1 replaced by a simpler solution. One reader, Saad Hassan also showed that in Q.5 in Paper 1, if we let $g(n)$ be the number of permutations σ of the set $\{1, 2, \ldots, n\}$ which satisfy the condition that $|\sigma(i) - i| \leq 1$ for all $i = 1, 2, \ldots, n$, then $g(n)$ satisfies a Finonacci relation. This has been added.

These commentaries started as an outgrowth of the author's annual commentaries on the Mathematics sections of the JEE Advanced Papers which began in 2003. In the commentary on 2017 JEE, while commenting on the declining quality of the JEE questions over the years, especially after the JEE became totally Multiple Choice Questions format, a comparison was drawn with the 2016 B.Sc. Entrance test of ISI (Indian Statistical Institute) which had some imaginatively designed problems even within a limited scope of the syllabus. Then to bring home the suggestion how the JEE could take cue from ISI Entrance tests, commentaries on the 2016 and subsequently the 2014 ISI Papers were completed and displayed.

The quality of the questions in these two years had raised expectations. But the first paper of 2015 turned out to be very disappointing. Also, a look at the question papers prior to 2014 revealed certain common patterns which were repeated. So the element of novelty was not so strong as it was when the commentaries for 2016 and 2014 were written. To some extent such repetitions are to be expected, given the limited syllabus and the large number of questions to be asked. Instead of asking 30 questions in Paper 1 to be solved in two hours, it would be far better to ask only about 15 really good questions.

As a result, the plan to write exhaustive commentaries on ISI Entrance Tests was curtailed to only the recent ones, viz. from 2014 onwards. So far, the commentaries from 2014 to 2017 ISI B.Sc. Entrance tests of ISI have been displayed. The present one is a sequel for the year 2018.

As in the other commentaries, unless otherwise stated, all the references are to the author's *Educative JEE Mathematics*, published by Universities Press, Hyderabad.

I am thankful to the persons who made the question papers available to me as well as those who pointed out errors in the earlier draft. I am especially thankful to Dr. Shailesh Shiral for the graph of $y = \frac{\sin x}{x}$ in Q. 19 of Paper 1.

Readers are invited to send their comments and point out errors, if any, in the commentary. They may be sent either by e-mail (kdjoshi314@gmail.com) or by an SMS or a WhatsApp message on mobile (9819961036).

ISI BStat-BMath-UGA-2018 Paper 1 with Comments

N.B. Each question has four options of which ONLY ONE is correct. There are 4 marks for a correctly answered, 0 marks for an incorrectly answered question and 1 mark for each unattempted question.

Q.1 Let $0 < x < \frac{1}{6}$ be a real number. When a certain biased dice is rolled, a particular face F occurs with probability $\frac{1}{6} - x$ and its opposite face occurs with probability $\frac{1}{6} + x$. Recall that the opposite faces sum to 7 in any dice. Assume that the probability of obtaining the sum 7 when two such dice are rolled is $\frac{13}{96}$. Then the value of x is

(A) $\frac{1}{8}$ (B) $\frac{1}{12}$ (C) $\frac{1}{24}$ (D) $\frac{1}{27}$.

Answer and Comments: (A). Denote the face opposite to F by F^*. Let A, A^* and B, B^* be the other two pairs of opposite faces. When two such dice are rolled, the possible outcomes are all 36 ordered pairs of the form (X, Y) where X, Y are (unrelated) faces of the dice. But these 36 pairs are not equally likely as the dice is biased. We are given

$$P(E) = \frac{13}{96} \tag{1}$$

where E is the event that the two faces X, Y are opposite to each other. This can happen in 6 possible ways: $(A, A^*), (A^*, A), (B, B^*), (B^*, B)$, (F, F^*) and (F^*, F). The probabilities of the first four cases are $\frac{1}{36}$ each. But those of the last two are $(\frac{1}{6} + x)(\frac{1}{6} - x)$ each. As these possibilities are mutually exclusive, (1) gives an equation in x, viz.

$$\frac{1}{9} + 2(\frac{1}{36} - x^2) = \frac{13}{96} \tag{2}$$

Simplifying,

$$\frac{1}{36} - x^2 = \frac{1}{2}(\frac{13}{96} - \frac{1}{9}) = \frac{1}{2} \times \frac{39 - 32}{288} = \frac{7}{576} \tag{3}$$

Hence

$$x^2 = \frac{1}{36} - \frac{7}{576} = \frac{16 - 7}{576} = \frac{9}{576} = \frac{1}{64} \tag{4}$$

which gives $x = \frac{1}{8}$. ∎

A good, simple problem on probability. The calculation involved is a bit laborious and prone to error. But the numerical data is so adjusted that x^2 is the square of a rational. This will serve as a warning in case some calculation goes wrong. The word 'biased' in the statement of the problem is a general word which conveys the opposite of 'fair'. It applies to other devices such as coins too. But in the case of a dice, when all the faces are not equally likely to appear, the dice is often called a **loaded dice**, because one way to design such a dice is to make the face opposite to the one you prefer to show slightly heavier than the others.

Q.2 An office has 8 officers including two who are twins. Two teams, Red and Blue, of 4 officers each are to be formed randomly. What is the probability that the twins would be together in the Red team?

(A) $\frac{1}{6}$ (B) $\frac{3}{7}$ (C) $\frac{1}{4}$ (D) $\frac{3}{14}$

Answer and Comments: (D). Another probability problem, but not as exciting as the last one. The Red team can be formed in $\binom{8}{4}$ i.e. in $\dfrac{8.7.6.5}{24} = 70$ ways. If both the twins are in it, then the remaining two officers can be chosen in $\binom{6}{2} = 15$ ways. So the desired probability is $\frac{15}{70} = \frac{3}{14}$. ∎

Q.3 Suppose Roger has 4 identical green tennis balls and 5 identical red tennis balls. In how many ways can Roger arrange these 9 balls in a line so that no two green balls are next to each other and no three red balls are together?

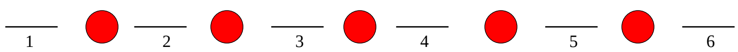
(A) 8 (B) 9 (C) 11 (D) 12

Answer and Comments: (D). A little narrative in the statement of a problem often serves an important task of testing a candidate's ability to isolate the mathematical essence of a problem from a real life situation. There is a distinct (and an alarmingly large) class of students who simply dread what they call 'word problems'. The reason is simple. In a word problem, nobody tells you what to do. You have to figure it out yourself! That is the very first step of applied mathematics.

Even when the conversion of a word problem to a mathematical one is fairly obvious, a little narrative can sometimes be refreshing. Sometimes it might make an oblique reference to some popular figure or sometimes the answer to the problem may come out to be consistent with some convention or superstition associated with the narrative (as when, in some problem, the ward number of a patient who did not survive an operation comes out to be 13). And, occasionally, the narrative may have a subtle humour which comes to surface when you find the answer. In the classic Sanskrit treatise *Leelavatee* on arithmetic by Bhaskaracharya, one of the problems asks to find how much money a miser gave to a beggar as alms. There is nothing great about the problem or the calculations *per se*. But the answer comes out to be the smallest monetary unit at that time. Had it come out higher, the miser would have to hide his face in shame!

But in the present problem, the narrative is serving no purpose. In essence the problem asks for the number of all possible arrangements of two kinds of objects, satisfying some restrictions. That these objects are tennis balls has nothing to do with the essence of the problem. And the introduction of the name Roger is absolutely high schoolish (actually, primary schoolish!). When the problem involves two (or more) persons, one can understand giving them some sweet names instead of the drab ones like A and B or X and Y. But in the present case, Roger has no role. (The only conceivable link is that one of the paper setters is a fan of the tennis great Roger Federer.) Thank God, the paper-setters did not further specify that Roger got the red tennis balls from his father and the green ones from his mother, in an attempt to make the problem more appealing!

Getting down to the mathematical part of the problem, problems of counting the number of linear arrangements of various types of objects so that no two objects of a particular type are together are very popular and there is a standard method to approach them. In the present problem, suppose first that the restriction is merely that no two green balls be adjacent. Then we first arrange the 5 red balls in a row leaving gaps in between two adjacent ones and also two gaps at the two ends where we can put the green balls as shown in the figure below.



The four green balls are to be placed into these six gaps. If no two green balls are to be next to each other, no gap can contain more than one green ball. So the number of possible arrangements is the same as the number of choosing 4 out of these 6 gaps and putting one green ball in each one of them (leaving the other two vacant). This number is $\binom{6}{4} = 15$.

But we have to exclude those arrangements in which three (or more) red balls are together, with no intervening green ball. For example, we must not leave gaps numbered 2 and 3 both vacant. Similarly, 3 and 4 cannot be both vacant and 4 and 5 cannot be both vacant. (It is all right if 1 and 2 are both vacant or 5 and 6 are both vacant.) So, from the 15 possible arrangements we got earlier, we have to exclude these three. Hence the answer is $15 - 3 = 12$. ∎

A fairly standard problem about arrangements where no two objects of a particular type are to appear together. If no two red balls are to be together too, then there is only one arrangement, in which the gaps numbered 2 to 5 are filled with one green ball each. More generally, if there are m red and n green balls and no two balls of the same colour are to be next to each other, then the number of such arrangements is 0 if $|m - n| > 1$, 1 if $|m - n| = 1$ and 2 if $|m - n| = 0$ (i.e. if $m = n$).

The problem gets very complicated if there are three or more types of objects. Suppose for example, that we have m red, n green and p blue balls and they are to arranged in a row with no two balls of the same colour next to each other. Although special cases can be worked out, there is no general formula as a function of m, n and p.

Q.4 The number of permutations σ of 1, 2, 3, 4 such that $|\sigma(i) - i| < 2$ for every $i = 1, 2, 3, 4$ is

(A) 2 (B) 3 (C) 4 (D) 5.

Answer and Comments: (D). In all there are $4! = 24$ permutations of the set $\{1, 2, 3, 4\}$. We can take them one-by-one and see how many of them meet the requirement. But this approach is bad for two reasons. First, even though 4 is a small number, 4! is not. In fact, as n grows, $n!$ grows very fast, (faster than, say, 10^n), so even for a relatively small value of n, a method based on an exhaustive checking all possible $n!$ permutations is exceedingly inefficient. Secondly, this approach would also require a systematic ordering of all $n!$ permutations lest we miss out some of them. Although such an ordering is possible, that will make the problem too complicated.

Another approach is to use that even though 4! is large, 4 is a small number and 2 is even smaller. So, the restriction that $|\sigma(i) - i| < 2$ means, simply that $\sigma(i) = i-1, i$ or $i+1$. So we can count the desired permutations $\sigma : \{1,2,3,4\} \longrightarrow \{1,2,3,4\}$ keeping track of these three possibilities. For $i = 1$, there are only two possibilities, either $\sigma(1) = 1$ or $\sigma(1) = 2$. If $\sigma(1) = 1$, then 1 is a fixed point of σ and σ induces a permutation of the set $\{2,3,4\}$. As this has 1 less element that the original set, perhaps an inductive approach will work. On the other hand, suppose that $\sigma(1) = 2$. Then $\sigma(2)$ cannot be 2 and so has to be either 1 or 3. If $\sigma(2) = 1$, then σ interchanges 1 and 2 and induces a permutation of the set $\{3,4\}$ which is a smaller set. So again an inductive approach seems feasible. But what if $\sigma(2) = 3$ (along with $\sigma(1) = 2$)? We claim that this cannot happen. For, σ is a bijection. So there is some $i \in \{1,2,3,4\}$ such that $\sigma(i) = 1$. But $i \neq 1, 2$ since $\sigma(1) = 2$ and $\sigma(2) = 3$. So, i has to be either 3 or 4. But then $|\sigma(i) - i| = |1 - i| > 1$, a contradiction.

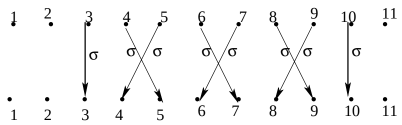
Summing up, we have shown that only two possibilities can arise for $\sigma(1)$. Either $\sigma(1) = 1$ and σ permutes the remaining elements $2, 3, 4$ among themselves or σ interchanges 1 and 2 and permutes the remaining elements $3, 4$ among themselves. In both the cases, the problem is reduced to a similar problem for a set with fewer than 4 elements and we can keep going.

We can now complete the solution by working out these tail ends in both the cases. But what we have learnt so far can be used to tackle a more general problem where 4 is replaced by any positive integer n (but the requirement $|\sigma(i) - i| \leq 1$ remains unchanged). So, suppose $\sigma : \{1, 2, \ldots, n\} \longrightarrow \{1, 2, \ldots, n\}$ is a bijection which satisfies $|\sigma(i) - i| \leq 1$ for all $i = 1, 2, \ldots, n$. We consider the fixed points of σ, i.e. points i for which $\sigma(i) = i$. σ may or may not have any fixed points. For example the permutation $\sigma = 2143$ (which, in the full form means $\sigma(1) = 2, \sigma(2) = 1, \sigma(3) = 4$ and $\sigma(4) = 3$) has no fixed points but satisfies the condition in the problem as it merely interchanges adjacent elements.

Now suppose j is any fixed point of σ, i.e. $\sigma(j) = j$. Let $A_j = \{1, 2, \ldots, j-1\}$ and $B_j = \{j+1, j+2, \ldots, n\}$. For $j = 1$, the subset A_j is empty, for $j = n$, the subset B_j is empty. But that makes no difference. The point is that every element of A_j differs from every element of B_j by at least 2. So σ cannot map an element of A_j to an element of B_j or vice versa. Figuratively, the fixed point j is a barrier which σ cannot cross. Hence σ permutes the elements of A_j within themselves and similarly permutes the elements of B_j within themselves.

Next consider the case where j and k are two fixed points of σ with $j < k$ and with no fixed point in between. Let $C = \{j+1, j+2, \ldots, k-1\}$. Again C may be empty if $k = j+1$. But by the same reasoning as above, σ cannot map any element of C to an element outside C or vice versa. So, σ induces a permutation of the subset C. We further show that this permutation must interchange adjacent pairs. Consider $j+1 \in C$. Then $j+2$ is also in C as otherwise $j+2 = k$, which would mean that $C = \{j+1\}$ and would make $j+1$ a fixed point of σ lying between j and k, contradicting the assumption. Also $\sigma(j+1) = j+2$ since $\sigma(j+1)$ can equal neither j nor $j+1$. (The latter possibility would make $j+1$ a fixed point of σ.) We now show that $\sigma(j+2) = j+1$. If this is not so, then since $j+2$ is not a fixed point, the only possibility would be $\sigma(j+2) = j+3$. But then the (unique) element, say s in C which maps to $j+1$ under σ cannot be any of $j, j+1$ and $j+2$, contradicting that $|\sigma(s) - s| \leq 1$. Thus we get that σ interchanges $j+1$ and $j+2$, i.e. $\sigma(j+1) = j+2$ and $\sigma(j+2) = j+1$. We now start with $j+3$ (in case it is in C), show that $\sigma(j+3) = j+4$ and thereafter $\sigma(j+4) = j+3$. In other words, σ interchanges $j+3$ and $j+4$. This goes on till we exhaust all the $k-j-1$ elements of C i.e. all elements between the two adjacent fixed points j and k of σ. In particular this also shows that $|C|$ i.e. $k-j$ is even (possibly 0).

The following diagram illustrates the case where $j = 3$ and $k = 10$. The arrows join i to $\sigma(i)$ for $3 \leq i \leq 10$.



The general picture should be clear now. Assume σ is a permutation of the set $\{1, 2, \ldots, n\}$ which satisfies the condition in the problem, viz. $|\sigma(i) - i| < 2$ for all i. Let $j_1, j_2, \ldots, j_r$ be the distinct fixed points of σ where $r \geq 0$ and $1 \leq j_1 < j_2 < \ldots j_r \leq n$. Then σ fixes all these elements. For notational uniformity, set $j_0 = 0$ and $j_{r+1} = n+1$. Then the number of elements between any j_p and j_{p+1} (where $0 \leq p \leq r$) is even (possibly 0) and σ interchanges these elements in pairs. In particular, this implies that r has the same parity as n, i.e. either both are even or both are odd. Moreover, j_p has the same parity as p, for $p = 0, 1, \ldots, r+1$. Hence the j's are alternately even and odd.

It follows that every σ is uniquely determined by these integers $0 = j_0 < j_1 < j_2 < \ldots < j_r < j_{r+1} = n+1$ which satisfy that j_p has the same parity as p for $p = 0, 1, 2, \ldots r, r+1$. Hence counting the number of such permutations amounts to finding how many sequences of this type there are. In the problem given, $n = 4$. So r can be either 4, 2 or 0. If $r = 4$, then σ is the identity permutation 1234 and we have $j_1 = 1, j_2 = 2, j_3 = 3$ and $j_4 = 4$. The next case is $r = 2$. Because of the requirement of alternating parity, we can have three possibilities (i) $j_1 = 1, j_2 = 2$, in which case $\sigma = 1243$, or (ii) $j_1 = 1, j_2 = 4$ in which case $\sigma = 1324$ and (iii) $j_1 = 3, j_2 = 4$, in which case $\sigma = 2134$. Finally, we have to consider the case where $r = 0$, i.e. σ has no fixed points. In this case $j_0 = 0, j_1 = 5$ and $\sigma = 2143$. Thus we see that in all there are five permutations of $\{1, 2, 3, 4\}$ of the desired type, viz. $1234, 1243, 1324, 2134$ and 2143. ∎

The answer in this special case, could, of course, have also been obtained by trial. But once the essential idea strikes (viz. that σ must permute the elements between two consecutive fixed points as adjacent pairs), the general case is equally easy. This problem, therefore, is more suited as a full length question (with a higher n than 4, say $n = 8$) in Paper 2. A special consequence of the reasoning above, which is interesting in its own right, is that if n is odd, then any permutation σ of $1, 2, \ldots, n$ which satisfies the condition $|\sigma(i) - i)| < 2$ for all $i = 1, 2, \ldots, n$, must have at least one fixed point. Even this special case would have been an interesting question for Paper 2.

Another interesting problem would be to let $g(n)$ be the number of all such permutations for a given n and ask to find a formula for $g(n)$

as a function of n. A direct answer here is not easy. But it is not hard to show that $g(n)$ satisfies a Fibonacci relation, viz.

$$g(n) = g(n-1) + g(n-2) \tag{1}$$

for all $n \geq 2$. This is easy to show. As we already noted, $\sigma(1)$ is either 1 or 2. In the first case the elements 2 to n are permuted among themselves and this can be done in $g(n-1)$ ways. In the second case, the elements 3 to n permute themselves in $g(n-2)$ ways. This proves (1).

However, the answer is not quite the n-th Fibonacci number F_n. These are defined recursively by $F_n = F_{n-1} + F_{n-2}$ with the initial conditions $F_0 = F_1 = 1$. In our problem the initial conditions are $g(1) = g(2) = 1$. So the answer is not $g(n) = F_n$ but $g(n) = F_{n+1}$. We skip the derivation of the formula for F_n. But using it the answer comes out to be

$$g(n) = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}\right] \tag{2}$$

Of course, one can *verify* (2) by induction on n, using the recurrence relation (1). Indeed that is essentially the JEE 1981 problem in Comment No. 11 of Chapter 4. In the problem that is asked here, we need only $g(4)$ which we calculated earlier as 5. This can also be derived from (2) by noting that

$$\begin{aligned} (1+\sqrt{5})^5 &= 1 + 5\sqrt{5} + 50 + 50\sqrt{5} + 125 + 25\sqrt{5} && (3)\\ \text{and} \quad (1-\sqrt{5})^4 &= 1 - 5\sqrt{5} + 50 - 50\sqrt{5} + 125 - 25\sqrt{5} && (4) \end{aligned}$$

Subtracting (4) from (3) and dividing the difference by $2^5\sqrt{5}$ i.e. by $32\sqrt{5}$ gives $g(4) = 5$.

Finally, we mention one possible extension of the problem which is really challenging. If σ satisfies $|\sigma(i) - i| = 0$ for every i, then it is the identity permutation. In the present problem the condition on σ is slightly relaxed. That is, $|\sigma(i) - i| \leq 1$ for every i. What if we relax it further to $|\sigma(i) - i| \leq 2$ for every i? Then counting the number of all permutations σ of $1, 2, \ldots, n$ which satisfy it will not be so easy.

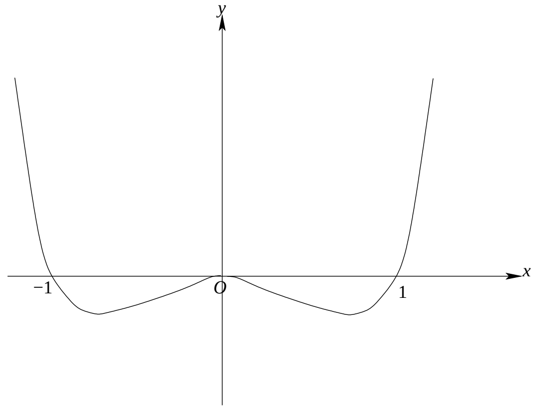
Q.5 Let $f(x)$ be a degree 4 polynomial with real coefficients. Let z be the number or real zeroes of f, and e be the number of local extrema (i.e. local maxima or minima) of f. Which of the following is a possible (z, e) pair?

(A) $(4, 4)$ (B) $(3, 3)$ (C) $(2, 2)$ (D) $(0, 0)$

Answer and Comments: (B). In general $f(x)$ has 4 roots. The complex roots, if any, appear in conjugate pairs. So the number of real roots is 4, 2 or 0. But in the first two cases, it is possible that two of these roots coincide. As for the local extrema, they always occur at the roots of $f'(x)$, which is a polynomial of degree 3. Hence e can be at most 3. This rules out (A). As for the remaining possibilities, we assume, without loss of generality that the leading coefficient of $f(x)$ is positive. If not, we replace f by $-f$. That does not affect the zeros of f. So z remains the same. Local maxima/mimima of $-f$ are the local minima/maxima of f respectively. So e also remains the same. Now (D) is easily ruled out because f has an absolute minimum over the the entire $I\!R$ (as its leading coefficient is positive and degree even) which is also a local minimum of f.

To see which of the remaining possibilities can hold, note that $f'(x)$ is a cubic polynomial. If $e = 2$, then $f'(x)$ has only two zeros, say a and b with $a < b$. One of these, say a, is a double root of $f'(x)$. But then f' has the same sign on both the sides of a (near a) and hence a cannot be a local extremum of f. That would mean $e = 1$, a contradiction. Hence (C) cannot hold. As only one of the options is correct, it has to be (B). ∎

The argument above presupposes that only one of the four options is correct. It is desirable to prove that (B) is indeed a possibility by giving an actual example. If $z = 3$, then f must have one double root, say b and two other simple roots, say a and c. The double root b is also a simple root of f' and hence a local extremum of f. The other two local extrema exist by Rolle's theorem, which says that in between any two zeros of f, there has to be a zero of f'. For an actual numerical example, take the double root as 0 and the other two roots of $f(x)$ as ± 1. Then $f(x) = x^2(x-1)(x+1) = x^4 - x^2$. $f'(x) = 4x^3 - 2x$ which vanishes at 0 and at $\pm\frac{1}{\sqrt{2}}$. So, f has three local extrema, viz. a local maximum at 0 and local minima at $\pm\frac{1}{\sqrt{2}}$. Hence (B) is an actual possibility. The graph of $y = f(x)$ is sketched below.



Chances are that some students have already seen graphs of this kind. They will immediately recognise that $z = 3, e = 3$ is a possibility. No further work is needed because the question does not ask to rule out the other possibilities logically, as we did above. So, the problem is a give-away for such students.

Q.6 A number is called a palindrome if it reads the same backward or forward. For example, 112211 is a palindrome. How many 6-digit palindromes are divisible by 495 ?

(A) 10 (B) 11 (C) 30 (D) 45

Answer and Comments: (B). Let the palindrome be $n = xyzzyx$. Since $495 = 11 \times 9 \times 5$, we have to check divisibility by 11, 9 and 5. As the sums of alternate digits are the same (viz. $x + y + z$), n is always divisible by 11. 5 divides n if and only if $x = 5$. (The value $x = 0$ is not allowed as the leading digit of n cannot be 0.) Finally, n will be divisible by 9 if and only if its digital sum $2(x + y + z)$ is divisible by 9, which reduces to $x + y + z$ being divisible by 9. So the problem amounts to counting all triples $(5, y, z)$ in which y, z can vary from 0 to 9 and $5 + y + z$ is a multiple of 9. Hence $y + z$ is either 4 or 13. The first case gives 5 possible solutions (with $y = 0, 1, 2, 3, 4$). In the second case, the possibilities are $y = 9, 8, 7, 6, 5, 4$. Hence in all there are 11 palindromes of the given type. ∎

A simple counting problem involving very elementary number theory (specifically, the criteria for divisibility by 11, 9 and 5). Perhaps the idea was merely to test a candidate's ability to understand a newly defined term. In that case, the given illustration is not sufficiently representative as it may give the wrong impression that the digits must be repeated in pairs. Better illustrations would be the numbers 305503 or 282282. Also it would have been better to include an illustration of a number which is not a palindrome, e.g. 370037. Anyway, the concept, if not the name, is familiar to most people through recreational words, such as ANNA or MALAYALAM or the short sentence 'Madam, I'm Adam' and, last but not least, the famous sentence attributed to Napoleon 'Able was I ere I saw Elba'.

Q.7 Let A be a square matrix of real numbers such that $A^4 = A$. Which of the following is true for every such A?

(A) det $(A) \neq -1$.
(B) A must be invertible.
(C) A cannot be invertible.
(D) $A^2 + A + I = 0$ where I is the identity matrix.

Answer and Comments: (A). The zero matrix satisfies the hypothesis but fails both (B) and (D). Also taking A to be the identity matrix, we see that (C) is false. That leaves (A) as the only correct option. However, to really justify it positively (rather than by elimination), repeatedly applying the multiplicativity property of determinants to the hypothesis, we get det $(A^4) = (\det(A))^4 =$ det (A). Clearly, this cannot hold if det $(A) = -1$. ∎

Whichever way you look at it, the problem is very simple. Option (D) is probably based on the factorisation of $A^4 - A$ as $A(A^3 - I)$ and further as $A(A-I)(A^2+A+I)$. If both A and $A-I$ are non-singular, then from $A^4 = A$, it would follow that $A^2 + A + I = 0$. Anyway, that is not applicable here. Perhaps the idea is to penalise those who know too much but do not know where not to apply what they know!

Lest the problem appears vacuous, we should give a non-trivial example where the hypothesis holds. One such source is what are called **idempotent matrices**. By definition, an idempotent matrix is a square matrix A for which $A^2 = A$. It then follows that A^3, A^4, and, in fact, all powers of A equal A (whence the name). The zero matrix and the identity matrix are idempotent. But any square matrix whose all non-diagonal entries are 0 and the diagonal entries are either 0 or 1 is also idempotent. A less trivial example is the 2×2 matrix $\begin{bmatrix} \cos^2\alpha & \sin\alpha\cos\alpha \\ \sin\alpha\cos\alpha & \sin^2\alpha \end{bmatrix}$ where α is any angle. (This matrix represents the projection of the plane $I\!R^2$ onto the line $y = (\tan\alpha)x$, which makes it easier to see why it is idempotent, without actually calculating its square.)

The matrix $A = \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix}$ also satisfies the hypothesis. It represents the complex number ω and satisfies $A^3 = I$.

Q.8 Consider the real valued function $h : \{0, 1, 2, \ldots, 100\} \longrightarrow I\!R$ such that $h(0) = 5, h(100) = 20$ and satisfying $h(i) = \frac{1}{2}(h(i+1) + h(i-1))$, for every $i = 1, 2, \ldots, 99$. Then the value of $h(1)$ is:

(A) 5.15 (B) 5.5 (C) 6 (D) 6.15.

Answer and Comments: (A). This is an example of a problem which is cunningly formulated. Once you recognise what its hypothesis means, the problem becomes trivial. The given relation can be rewritten as

$$2h(i) = h(i-1) + h(i+1) \tag{1}$$

and hence as

$$h(i+1) - h(i) = h(i) - h(i-1) \tag{2}$$

for every $i \geq 1$. From (2), $h(0), h(1), h(2), \ldots, h(100)$ is an airthmetic progression. (Of course, this can also be seen without such a rewriting, because the given condition means that every term is the arithmetic mean of the two terms besides it, which is just another way of looking at an A.P.)

After this there is hardly anything left in the problem. If d is the common difference of the A.P. then we have

$$h(100) = h(0) + 100d \tag{3}$$

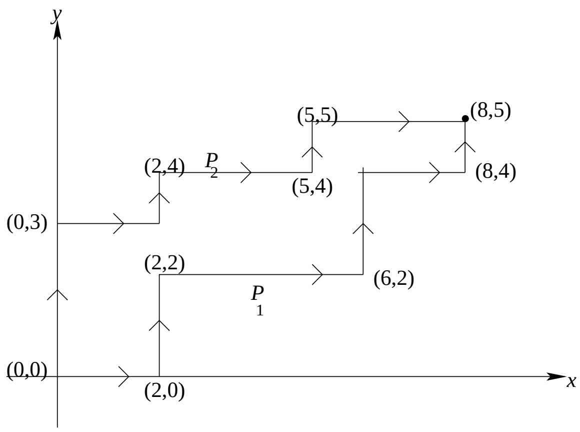
From $h(0) = 5$ and $h(100) = 20$ we get $100d = 20 - 5 = 15$. Hence $d = 0.15$. So, $h(1) = h(0) + d = 5 + 0.15 = 5.15$. ∎

There is not much to comment as mathematically the problem is trivial. Such questions are more suitable in live interviews to test the mental alertness of a candidate.

Q.9 An up-right path is a sequence of points $a_0 = (x_0, y_0), a_1 = (x_1, y_1), a_2 = (x_2, y_2), \ldots$ such that $a_{i+1} - a_i$ is either $(1, 0)$ or $(0, 1)$. The number of up-right paths from $(0, 0)$ to $(100, 100)$ which pass through $(1, 2)$ is:

(A) $3.\binom{197}{99}$ (B) $3.\binom{100}{50}$ (C) $2.\binom{197}{98}$ (D) $3.\binom{197}{100}$.

Answer and Comments: (A). Yet another problem where the essential idea is very simple but the wording is (perhaps intentionally) clumsy. It would be much easier to describe an up-right path as a sequence of unit steps, where each unit step is either in the direction of the positive x-axis or in the direction of the positive y-axis. (Customarily, these directions are taken as east and north respectively. So an up-right path can also be called a north-east path, in which every unit step is either northward or eastward. If we denote each unit east step by H (for horizontal) and each unit north step by V (for vertical), then every up-right path is determined by a unique sequence of these H's and V's. (Such sequences are called **binary sequences**.)



In the diagram above we show two up-right paths, P_1 and P_2, from $(0, 0)$ to $(8, 5)$. Each path has 13 unit steps, of which 8 are H

and 5 are V. The binary sequence corresponding to the path P_1 is $(H, H, V, V, H, H, H, H, V, V, H, H, V)$ while that corresponding to the path P_2 is $(V, V, V, H, H, V, H, H, H, V, H, H, H)$.

Given a point (a, b) and positive integers m, n it is clear that every up-right path from (a, b) to $(a + m, b + n)$ corresponds to a unique binary sequence of length $m+n$ of V's and H's in which m entries are H and the remaining n entries are V. So the total number of all such paths is $\binom{m+n}{m}$ (which is the same as $\binom{m+n}{n}$).

In the given problem, we first count the number of up-right paths from $(0, 0)$ to $(1, 2)$. We can follow any of these paths by any up-right path from $(1, 2)$ to $(100, 100)$. There are $\binom{3}{1} = 3$ paths of the first type and $\binom{197}{99}$ paths of the second type. So the total number of up-right paths from $(0, 0)$ to $(100, 100)$ that pass through $(1, 2)$ is $3.\binom{197}{99}$. ∎

An easy problem once the essential idea is understood.

Q.10 Let $f(x) = \frac{1}{2}x \sin x - (1 - \cos x)$. The smallest positive integer k such that $\lim_{x \to 0} \frac{f(x)}{x^k} \neq 0$ is:

(A) 3 (B) 4 (C) 5 (D) 6.

Answer and Comments: (B). Clearly $f(x) \to 0$ as $x \to 0$. Also, for every positive integer k, $x^k \to 0$ as $x \to 0$. So three things are possible for the behaviour of the ratio $\frac{f(x)}{x^k}$ as $x \to 0$. For small k, $f(x)$ tends to 0 more rapidly than x^k and so the ratio tends to 0. For large k, x^k tends to 0 faster than $f(x)$ and then the limit of the ratio does not exist. But there is a critical value of k for which x^k tends to 0 as rapidly as $f(x)$ and then the ratio $\frac{f(x)}{x^k}$ tends to a finite non-zero number as x tends to 0. In such a case we say that near 0, $f(x)$ is **comparable to** x^k. The problem asks to determine which power of x is comparable to the given function $f(x)$ as $x \to 0$.

The basic result needed is that as $x \to 0$, $\sin x$ is comparable to x. This follows from the well known result that

$$\lim_{x \to 0} \frac{\sin x}{x} = 1 \tag{1}$$

Let us write $f(x)$ as $g(x) - h(x)$ where

$$\begin{aligned} g(x) &= \frac{1}{2}x\sin x & (2)\\ \text{and} \quad h(x) &= 1-\cos x = 2\sin^2(\frac{x}{2}) & (3)\end{aligned}$$

Because of (1), both $g(x)$ and $h(x)$ are comparable to x^2 as $x \to 0$. So it is tempting to think that the answer is $k = 2$. But there is a catch here. Let L_1 and L_2 be, respectively, the limits of the ratios $\frac{g(x)}{x^2}$ and $\frac{h(x)}{x^2}$ as $x \to 0$. Both are non-zero. If $L_1 \neq L_2$, then $\frac{g(x)-h(x)}{x^2}$ will tend to $L_1 - L_2$ and then k would be 2. But if both these limits happen to be equal then $\frac{f(x)}{x^2}$ tends to 0 as $x \to 0$. In such a case, $f(x)$ is not comparable to x^2 but to some higher power of x. By not giving 2 as a possible answer, the paper-setters have given an implicit hint that $L_1 = L_2$. We can, of course, verify this because from (2), both L_1 and L_2 equal $\frac{1}{2}$.

This makes the problem more delicate. We now have to work with $\lim\limits_{x\to 0} \frac{\frac{1}{2}x\sin x - 2\sin^2(\frac{x}{2})}{x^k}$ and decide for which value of k this limit is finite and non-zero. As this limit is of the indeterminate $\frac{0}{0}$ form, one method is to try l'Hôpital's rule. Since derivatives will be involved, it is convenient to put back $f(x)$ in its original form as $\frac{1}{2}x\sin x - 1 + \cos x$. Two applications of the rule give

$$\begin{aligned} \lim_{x\to 0}\frac{f(x)}{x^k} &= \lim_{x\to 0}\frac{\frac{1}{2}x\sin x - 1 + \cos x}{x^k} \\ &= \lim_{x\to 0}\frac{\frac{1}{2}\sin x + \frac{1}{2}x\cos x - \sin x}{kx^{k-1}} \\ &= \lim_{x\to 0}\frac{-\frac{1}{2}\cos x - \frac{1}{2}x\sin x + \frac{1}{2}\cos x}{k(k-1)x^{k-2}} & (4)\\ &= \lim_{x\to 0}\frac{-\frac{1}{2}x\sin x}{k(k-1)x^{k-2}} & (5)\end{aligned}$$

where, in going from (4) to (5) we have not applied the l'Hôpital's rule, but merely simplified the numerator. We can go on applying the rule again. But that is hardly necessary. We already know that as $x \to 0$,

$x \sin x$ is comparable to x^2. So, if $k = 4$, then the limit in (4) is a finite, non-zero number. The actual value of this limit can be found as $\dfrac{-1/2}{4 \times 3}$. But that is not necessary. All that matters is that it is non-zero. Hence 4 is the least value of the integer k for which the given limit is non-zero. (Actually, that is an understatement, because 4 is the *only* value of k for which the limit is non-zero. For $k < 4$, the limit is 0 while for $k > 4$ it does not exist.) ∎

A good problem based on the concept of comparable rates of growth. There is also a solution based on the Taylor series expansions of $\sin x$ and $\cos x$. Using these, we get

$$\begin{aligned} f(x) &= \frac{x}{2}(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots) - 1 + (1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots) \\ &= \frac{x^2}{2} - \frac{x^4}{2.3!} + \frac{x^6}{2.5!} + \ldots - \frac{x^2}{2} + \frac{x^4}{4!} - \ldots \\ &= A_4 x^4 + A_6 x^6 + \ldots \end{aligned} \tag{6}$$

where $A_4 = \frac{1}{24} - \frac{1}{12} = -\frac{1}{24} \neq 0$. As $x \to 0$, higher powers of x are negligible and so $f(x)$ is comparable to x^4 as $x \to 0$. This solution is too advanced for the HSC level. But the series for $\sin x$ and $\cos x$ are very popular. And, in any case, for a multiple choice format question, it does not matter how you get the answer, as long as it is correct.

Q.11 Nine students in a class gave a test for 50 marks. Let $S_1 \leq S_2 \leq \ldots \leq S_5 \leq \ldots \leq S_8 \leq S_9$ denote their ordered scores. Given that $S_1 = 20$ and $\sum\limits_{i=1}^{9} S_i = 250$, let m be the smallest value that S_5 can take and let M be the largest value that S_5 can take. Then the pair (m, M) is given by

(A) $(20, 35)$ (B) $(20, 34)$ (C) $(25, 34)$ (D) $(25, 50)$.

Answer and Comments: (B). From the data, $S_5 \geq 20$. If $S_5 = 20$, then S_2, S_3, S_4 also equal 20 and so $S_6 + S_7 + S_8 + S_9 = 250 - 100 = 150$. This is feasible if we take $S_8 = S_9 = 50$ and $S_6 = S_7 = 25$. So $m = 20$. To determine M, note that if $S_5 = M$ then S_6, S_7, S_8, S_9 are at least M each and so $S_5 + S_6 + S_7 + S_8 + S_9 \geq 5M$. But $S_1 + S_2 + S_3 + S_4 \geq 80$.

So we have $250 \geq 5M + 80$ which gives $M \leq 34$. Again this is feasible if we let $S_i = 20$ for $i = 1, 2, 3, 4$ and $S_i = 34$ for $i = 5, 6, 7, 8, 9$. So $M = 34$. ∎

A simple problem about inequalities. It is not clear what role the maximum score in the test (viz. 50) has. Had it been, say 30, the problem would be more interesting.

Q.12 Let 10 red balls and 10 white balls be arranged in a straight line such that 10 each are on either side of a central mark. The number of such symmetrical arrangements about the central mark is

(A) $\frac{10!}{5!5!}$ (B) 10! (C) $\frac{10!}{5!}$ (D) 2.10!

Answer and Comments: (A). Because of symmetry, on each side there will be 5 red and 5 white balls. These can be arranged in $\frac{10!}{5!5!}$ ways. Once they are arranged, the arrangement on the other side of the mark is determied uniquely by symmetry. ∎

An absolutely simple problem. The basic idea is that of a palindrome in Q. 6. But there the subsequent work involved some number theory. Here there is hardly anything left except arranging 5 red and 5 white balls in a row, for which there is a formula.

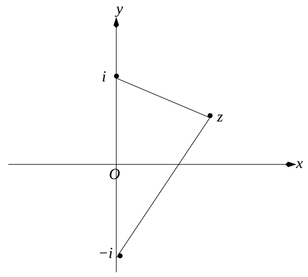
Q.13 If $z = x + iy$ is a complex number such that $\left|\frac{z-i}{z+i}\right| < 1$, then we must have

(A) $x > 0$ (B) $x < 0$ (C) $y > 0$ (D) $y < 0$.

Answer and Comments: (C). One way to tackle the problem would be to convert the data in terms of x and y. But usually, it is better to work in terms of complex numbers. The given inequality can be rewritten as

$$|z - i| < |z + i| \tag{1}$$

The L.H.S. is the distance of the point z from the point i while the R.H.S. is its distance from $-i$. So, (1) is equivalent to saying that the point z is closer to i than to $-i$. Therefore z lies on that side of the perpendicular bisector of the segment joining i to $-i$, which contains i. The perpendicular bisecor here is simply the x-axis. So, (1) holds if and only if $y > 0$. ∎



An easy problem if approached geometrically. Just in case, you can't think of it, we give the approach based on conversion to real numbers. Then squaring both the sides of (1) we get

$$x^2 + (y-1)^2 < x^2 + (y+1)^2 \tag{2}$$

which is equivalent to

$$-2y < 2y \tag{3}$$

i.e. to $y > 0$. So, this approach, at least in the present problem, is almost equally easy.

Q.14 Let $S = \{x - y \mid x, y \text{ are real numbers with } x^2 + y^2 = 1\}$. Then the maximum number in the set S is

(A) 1 (B) $\sqrt{2}$ (C) $2\sqrt{2}$ (D) $1 + \sqrt{2}$.

Answer and Comments: (B). This question resembles Q.8 in that once the clumsy formulation is unveiled, the problem becomes very familiar. Taking $x = \cos\theta$ and $y = \sin\theta$, the problem asks to maximise $\cos\theta - \sin\theta$ as θ varies over $[-\pi, \pi]$. Instead of applying calculus, we rewrite $\cos\theta - \sin\theta$ as $\sqrt{2}(\cos\theta \sin(\frac{\pi}{4}) - \sin\theta\cos(\frac{\pi}{4})$ and further as $\sqrt{2}\sin(\frac{\pi}{4} - \theta)$. Its maximum occurs when $\theta = -\frac{\pi}{4}$. And the maximum value is $\sqrt{2}$. (If we let θ vary over $[0, 2\pi]$, which is more customary, the maximum occurs when $\theta = \frac{7\pi}{4}$. But it gives the same point.) ∎

Q.15 In a factory, 20 workers start working on a project of packing consignments. They need exactly 5 hours to pack one consignment. Every hour 4 new workers join the existing workforce. It is mandatory to relieve a worker after 10 hours. Then the number of consignments that would be packed in the initial 113 hours is

(A) 40 (B) 50 (C) 45 (D) 52.

Answer and Comments: (C). There is some ambiguity regarding the type of work involved. Packages are discrete objects. And one tends to expect that when 20 workers are working on a particular package, they will be working as a team and till they finish packaging one consignment, it is of no use to add to the team. So the new members who join will have some work to do only when they reach a strength of 20 and then they will work on a separate consignment.

But if we assume this interpretation, then the new workers who join will be idle (or 'sitting on the bench' in the common parlance of the IT industry!) for considerable time. The mandatory relief, however, has to be given to a worker 10 hours after joining, rather than after ten hours of working. The resulting wastage is considerable and makes this interpretation unrealistic.

So, we assume that the packaging work is not discrete but continuous in nature and also that every worker starts contributing independently immediately upon joining. Each consignment takes 100 man-hours of work. But these can be contributed by any number of workers. As a real life example, take the work of painting walls. If 20 workers can paint a wall in 5 hours, then 4 workers working together at any moment will do it in 25 hours, with the understanding that they may not be the same four workers all the time. (A more realistic model will be the job of filling water tanks of capacity 100 units each, by pumps each of which pumps 1 unit of water per hour and has to be stopped after operating for 10 hours. Any number of pumps can simultaneously pump water into the same tank and as soon as one tank is full, all the pumps filling into it are instantaneously shifted to start filling another.)

With this interpretation, the problem resembles the umpteeen number of problems we do in school regarding time, work and rate. All we need to do is to calculate the total man-hours accumulated in the first 113 hours and divide this number by 100. The real problem is with this calculation, because not only the workers keep changing, their numbers also keep changing.

We begin by keeping an hourly track of how many workers are working. In the first hour, there are 20 workers. In the second there are 24, in the third there are 28 and so on. This goes on till the tenth hour during which we have 56 workers. At the end of the tenth hour, 4 new workers will join but the the original 20 workers will be relieved. Hence during the 11-th hour, there will be only 40 workers. After this there will be a steady state, because at the end of each hour, as many new workers will join as will be relieved. So from the 11-th hour to the 113-th hour, at any time 40 workers will be working.

If we denote the total number of man hours during the first 113 hours by W, we get

$$\begin{aligned} W &= 20 + 24 + 28 + \ldots + 56 + (113 - 10) \times 40 && (1) \\ &= 380 + 4120 && (2) \\ &= 4500 && (3) \end{aligned}$$

where in going from (1) to (2) we have used the formula for the sum of the terms in an A.P. (We can do without it too, by adding terms which are symmetrically located, like $20 + 56$, $24 + 52$, till $36 + 40$.)

Since each packaging consignment takes 100 man-hours, from (3) we see that 45 consignments would be completed during the first 113 hours. ∎

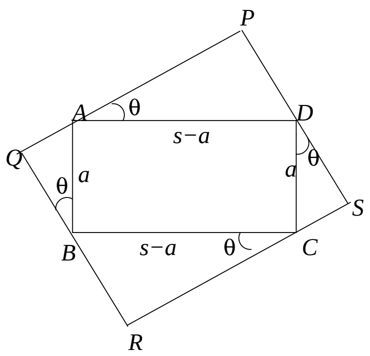
As in many earlier problems, the mathematical essence of the problem is very elementary. Such problems are, therefore, more a test of a candidate's ability to reduce a real life problem to a mathematical one. One only wishes that the formulation were free of ambiguity.

Q.16 Let $ABCD$ be a rectangle with its shorter side $a > 0$ units and perimeter $2s$ units. Let $PQRS$ be a rectangle such that the vertices A, B, C and D respectively lie on the lines PQ, QR, RS and SP. Then the maximum area of such a rectangle $PQRS$ in square units is given by

(A) s^2 (B) $2a(s - a)$ (C) $\frac{s^2}{2}$ (D) $\frac{5}{2}a(s - a)$.

Answer and Comments: (C). The manner in which the data is given is somewhat puzzling. Normally, the size of a rectangle is specified in terms of its length and the breadth, say b and a (since usually the breadth is smaller than the length). Then its semi-perimeter is $s = a+b$. Mathematically, if s and a are given one can determine b and so it makes no difference which two of the three variables a, b, s are specified. The unusual choice of a and s, rather than of a and b, suggests that the correct answer is in terms of s. So perhaps (A) or (C) is the right answer.

Mathematically, this guess has no validity. Nor does it really matter. With the data as given, the longer side of the rectangle $ABCD$ is $s-a$. We take AB as a and AD as $s-a$. Let θ be the angle between the lines AD and PQ. Then θ is also the angle between BC and RS, as they are parallel, respectively to AD and PQ. Moreover, θ is also the angle between AB and QR since $AB \perp AD$ and $QR \perp PQ$. Similarly, θ equals the angle between CD and PS, as shown in the diagram below.



We express the area, say Δ, of the rectangle $PQRS$ in terms of a, s and θ as follows.

From the figure above, we have

$$PQ = PA + AQ = AD\cos\theta + AB\sin\theta = (s-a)\cos\theta + a\sin\theta \quad (1)$$

and similarly,

$$PS = PD + DS = (s-a)\sin\theta + a\cos\theta \quad (2)$$

Hence

$$\begin{aligned}\Delta &= PQ.PS \\ &= a(s-a) + (a^2 + (s-a)^2)\sin\theta\cos\theta \qquad (3)\end{aligned}$$

Since a and $s-a$ are constants, Δ is a function of θ and it will be maximum if and only if $\sin\theta\cos\theta$ is maximum. Instead of finding the maximum by calculus, we express $\sin\theta\cos\theta$ as $\frac{1}{2}\sin 2\theta$ which is maximum when $\theta = \frac{\pi}{4}$. The maximum value of Δ is $a(s-a) + \frac{1}{2}(a^2 + (s-a)^2) = as - a^2 + a^2 + \frac{s^2}{2} - as = \frac{s^2}{2}$. ∎

A good problem, involving elementary geometry and trigonometry, but more importantly, the ability to visualise the data. It could have been asked as a full length question in Paper 2, along with a straight edge and compass construction for this rectangle with maximum area. A still more challenging problem would be to ask a construction for the rectangle $PQRS$ whose sides contain the vertices of a given rectangle $ABCD$ and whose area equals that of some other given rectangle $A'B'C'D'$. In that case one would first have to construct 2θ (which may be less than $\frac{\pi}{2}$) using (3). Bisecting it will give θ. After getting hold of θ, the point P can be determined as the intersection of the semi-circle with AD as the diameter and a line making angle θ with the side AB.

Q.17 The number of pairs of integers (x, y) satisfying the equation $xy(x+y+1) = 5^{2018} + 1$ is :

(A) 0 (B) 2 (C) 1009 (D) 2018.

Answer and Comments: (A). Equations in several variables where the variables can take only integral values are called **diophantine equations**. Usually, they involve higher powers of these variables and there is no general method to solve them. Each problem is in its own class. Even a slight difference can drastically affect the degree of difficulty. For example all solutions of the **P**ythagorean equation $x^2 + y^2 = z^2$ are known. Cancelling common factors, if any, (so that x, y, z are pairwise coprime), it can be shown that they are of the form $x = u^2 - v^2$ and $y = 2uv$ (or vice versa) for some integers u, v (so that $z = u^2 + v^2$). But the deceptively similar equation $x^3 + y^3 = z^3$ eluded as great mathematicians as Gauss and others for more than a century. **Fermat** conjectured that for $n > 2$, the diophantine equation $x^n + y^n = z^n$ has no solutions in positive integers. This conjecture was settled (in the affirmative) only recently. And the methods needed are far too advanced.

On this background, when a diophantine equation is given in an elementary examination, it is a safe bet to say that it will have either no solutions or only obvious solutions, and further, that the methods needed will generally not go beyond divisibility. The present problem is no exception. We claim that it has no solutions. And the key to the proof is the observation that the R.H.S. $5^{2018} + 1$ is divisible by 2 but not by 4. Divisibility by 2 is obvious since all powers of 5 are odd. For showing non-divisibility by 4, we note that the successive powers of 5 starting from 5^2 end in 25. For example, the first few such powers are $25, 125, 625, 3125, \ldots$. For an inductive proof, if $5^k = 100r + 25$, then $5^{k+1} = 5(100r + 25) = (5r + 1)100 + 25$.

As a result, $5^{2018} + 1$ is of the form $100m + 26$ for some integer m. Therefore it is divisible by 2 but not by 4. So, if $xy(x+y+1) = 5^{2018}+1$, then exactly one of the three factors x, y and $(x + y + 1)$ is even. But this can never happen. Surely, x, y are not both even. If one of them is even and the other odd, then the factor $x + y + 1$ would be even, a contradiction. The only case left is when both x, y are odd. But in that case $x + y + 1$ and hence $xy(x + y + 1)$ would be odd, a contradiction.

Hence the given equation has no solutions. ∎

Q.18 Let $p(n)$ be the number of digits when 8^n is written in base 6, and let $q(n)$ be the number of digits when 6^n is written in base 4. For example, 8^2 in base 6 is 144, hence $p(2) = 3$. Then $\lim\limits_{n\longrightarrow\infty} \dfrac{p(n)q(n)}{n^2}$ equals :

(A) 1 (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) 2.

Answer and Comments: (C). A k-digit number with base 6 is at least 6^{k-1} and at most $6^k - 1$. So if 8^n has $p(n)$ digits when expressed with base 6, we have

$$6^{p(n)-1} \leq 8^n < 6^{p(n)} \tag{1}$$

Taking logarithms, this means

$$(p(n) - 1)\log 6 \leq n \log 8 < p(n) \log 6 \tag{2}$$

This is valid regardless of the base of the logarithms, as long as the base is greater than 1. As we shall see, the actual base does not matter for our purpose. For the sake of definiteness, we assume that all logarithms are with base 10. Dividing throughout by $n \log 6$, we get

$$\frac{p(n) - 1}{n} \leq \frac{\log 8}{\log 6} < \frac{p(n)}{n} \tag{3}$$

which can be rewritten as

$$\frac{\log 8}{\log 6} < \frac{p(n)}{n} \leq \frac{\log 8}{\log 6} + \frac{1}{n} \tag{4}$$

for every positive integer n. We now apply the Sandwich Theorem for limits of sequences. The first term is a constant sequence while the last term is a sequence which tends to this constant (since $\frac{1}{n} \to 0$ as $n \to \infty$). So, we get

$$\lim_{n\to\infty} \frac{p(n)}{n} = \frac{\log 8}{\log 6} \tag{5}$$

By an entirely analogous argument, we have

$$\lim_{n\to\infty} \frac{q(n)}{n} = \frac{\log 6}{\log 4} \tag{6}$$

From (5) and (6),

$$\lim_{n\to\infty} \frac{p(n)q(n)}{n^2} = \frac{\log 8}{\log 4} \tag{7}$$

It only remains to evaluate $\frac{\log 8}{\log 4}$. Had we taken the base 2, the numerator and the denominator would be 3 and 2 respectively. Even for any other base, say 10, by the laws of logarithms, we have $\log 8 = 3 \log 2$ and $\log 4 = 2 \log 2$ and so the value of $\frac{\log 8}{\log 4}$ is $\frac{3}{2}$. ∎

An excellent and a well designed problem, based on very simple properties of logarithms. The solution looks longer because we have given it rigorously. In informal work, instead of (1) we can write that for large n,

$$8^n \approx 6^{p(n)} \tag{8}$$

and hence

$$n \log 8 \approx p(n) \log 6 \tag{9}$$

This gives a quick derivation of (5). (6) can be derived analogously.

Q.19 For a real number α, let S_α denote the set of those real numbers β that satisfy $\alpha \sin(\beta) = \beta \sin(\alpha)$. Then which of the following statements is true?

(A) For any α, S_α is an infinite set.

(B) S_α is a finite set if and only if α is not an integer multiple of π.

(C) There are infinitely many numbers α for which S_α is the set of all real numbers.

(D) S_α is always finite.

Answer and Comments: (B). Because of the periodicity of the trigonometric functions, for a given α, there are infinitely many angles β such that $\sin \beta = \sin \alpha$. In fact all such angles are given by

$$\beta = n\pi + (-1)^n \alpha \tag{1}$$

where n is any integer. But this knowledge is of little help because the equation in the present problem is different, viz.

$$\alpha \sin \beta = \beta \sin \alpha \tag{2}$$

For $\alpha, \beta \neq 0$, this can be recast as

$$\frac{\sin \beta}{\beta} = \frac{\sin \alpha}{\alpha} \tag{3}$$

or equivalently, as

$$f(\beta) = f(\alpha) \tag{4}$$

where the function $f(x)$ is defined by

$$f(x) = \begin{cases} \frac{\sin x}{x}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases} \tag{5}$$

Then f is an even function (i.e. $f(-x) = f(x)$ for all x) which is continuous everywhere. We are more interested in the behaviour of $f(x)$ as $x \to \pm\infty$. As the function is even, we confine ourselves only to $x > 0$. The numerator $\sin x$ is periodic and varies between -1 and 1 in any interval of length 2π. But the denominator x is not periodic. In fact, it keeps on increasing as $x \to \infty$ and for large x, $\sin x$ is insignificant as compared to x. Put differently,

$$\lim_{x\to\infty} f(x) = 0 \tag{6}$$

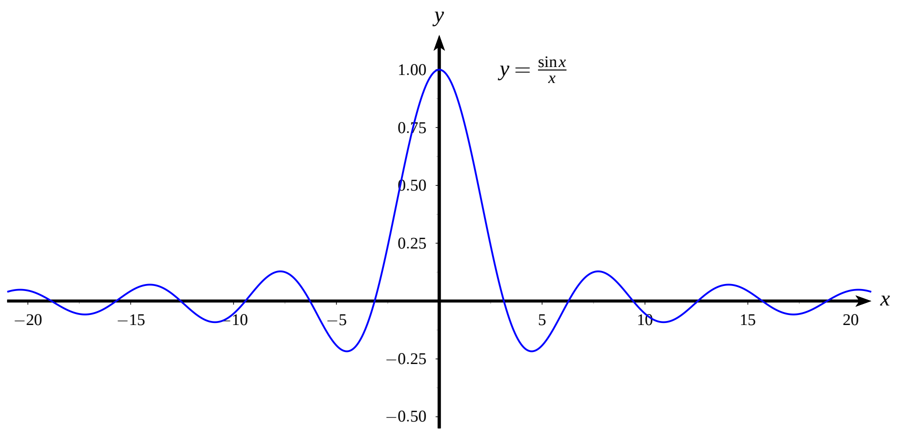
which follows because we have $|f(x)| = \frac{|\sin x|}{x} \leq \frac{1}{x}$ for all $x > 0$.

As with the sine function, the zeros of $f(x)$ are precisely all integral multiples of π. Also like the sine function, if n is even then $f(x) > 0$ for $x \in (n\pi, (n+1)\pi)$ and if n is odd, then $f(x) < 0$ if $x \in (n\pi, (n+1)\pi)$. So, the graph of $f(x)$ is also wavelike, like that of the sine function, but because of the factor x in the denominator, the 'amplitudes' of these waves go on decreasing. In fact, for $x \in [n\pi, (n+1)\pi]$, we have

$$|f(x)| \leq \frac{1}{n\pi} \tag{7}$$

which gives a ceiling on the amplitudes.

The graph of $y = f(x)$ is shown below.



In terms of the function f, for every α, $S_\alpha = \{\beta : f(\beta) = f(\alpha)\}$. If $\alpha = n\pi$ for some $n \in \mathbb{Z}$, then $f(\alpha) = 0$ and there are infinitely many points in the set S_α since $f(n\pi) = 0$ for all $n \in \mathbb{Z}$. On the other hand, if α is not an integral multiple of π, then $f(\alpha) \neq 0$. Let $\epsilon = |f(\alpha)|$. Then $\epsilon > 0$ and so by (6), there exists some $R > 0$ such that for all $|x| > R$, $|f(x)| < \epsilon$. We can take R to be $k\pi$ for some sufficiently large integer k. Then the possible values of β for which $f(\beta) = f(\alpha)$ all lie in the interval $[-k\pi, k\pi]$. Moreover β cannot be of the form $m\pi$ for any $m \in \mathbb{Z}$. So every element of the set S_α lies in an open interval $(m\pi, (m+1)\pi)$ for some m, $-k \leq m < k$.

If we could show that each of these $2k$ open intervals contains at most three elements of S_α, it would follow that the set S_α is finite. This is obvious from the graph. (In fact, from the graph, each such interval contains at most two elements of S_α.) However, for an analytical proof, suppose there exist four elements, say, $\beta_1 < \beta_2 < \beta_3 < \beta_4$ of S_α in $(m\pi, (m+1)\pi)$, then each $f(\beta_i)$ equals $f(\alpha)$ and so by Rolle's theorem, there would be points $\gamma_1 \in (\beta_1, \beta_2), \gamma_2 \in (\beta_2, \beta_3)$ and $\gamma_3 \in (\beta_3, \beta_4)$ such that

$$f'(\gamma_1) = f'(\gamma_2) = f'(\gamma_3) = 0 \tag{8}$$

Now note that

$$f'(x) = \frac{x\cos x - \sin x}{x^2} \tag{9}$$

So, the zeros of f' are also the zeros of $g(x) = \tan x - x$ which is undefined at $x = \frac{2m+1}{2}\pi$. But on each of the two subintervals $[m\pi, \frac{2m+1}{2}\pi)$ and $(\frac{2m+1}{2}\pi, (m+1)\pi]$, $g(x)$ is a strictly increasing function on as we see from its derivative $\sec^2 x - 1$. So each of these two subintervals contains at most one zero of $\tan x - x$ and hence of $f'(x)$. But that contradicts (8) since at least two γ's must be in one of these two subintervals.

Summing up, we have shown that the set S_α is infinite if and only if $\sin\alpha = 0$, i.e. α is an integer multiple of π. This is precisely (B), in its contrapositive form. ∎

A good problem requiring study of the graph of $y = \frac{\sin x}{x}$. More suitable for Paper 2 where reasoning could be tested.

Q.20 If $A = \begin{pmatrix} 1 & 1 \\ 0 & i \end{pmatrix}$ and $A^{2018} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $a + d$ equals:

(A) $1 + i$ (B) 0 (C) 2 (D) 2^{2018}.

Answer and Comments: (B). The matrix A is an example of what is called an **upper triangular** matrix, i.e. a matrix in which all entries below the diagonal are 0. By a direct calculation, for any real (or complex) numbers $x, y, z, x', y,' z'$, we have

$$\begin{pmatrix} x & y \\ 0 & z \end{pmatrix} \begin{pmatrix} x' & y' \\ 0 & z' \end{pmatrix} = \begin{pmatrix} xx' & xy' + yz' \\ 0 & yy' \end{pmatrix} \tag{1}$$

Verbally, the product of two upper triangular matrices (of order 2) is an upper triangular matrix whose diagonal elements are the products of the respective diagonal elements of the two matrices.

Applying this fact repeatedly, we see that

$$A^{2018} = \begin{pmatrix} 1^{2018} & \lambda \\ 0 & i^{2018} \end{pmatrix} \tag{2}$$

for some complex number λ. Equating this with $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we get $a = 1^{2018} = 1$, $d = i^{2018} = i^2 = -1$. So $a + d = 1 + (-1) = 0$. ∎

A simple problem for those who know (1). Those who don't, can begin by calculating A^2 and A^3 by direct multiplication. Then they will guess that the diagonal entries of A^n are the n-th powers of the corresponding diagonal entries of A. A scrupulous student will foolishly waste time in proving this guess by induction on n. But ultimately, all will get the answer.

Q.21 Let $f : I\!\!R \longrightarrow I\!\!R$ and $g : I\!\!R \longrightarrow I\!\!R$ be two functions. Consider the following two statements.

P(1) : If $\lim_{x\to 0} f(x)$ exists and $\lim_{x\to 0} f(x)g(x)$ exists, then $\lim_{x\to 0} g(x)$ must exist.

P(2) : If f, g are differentiable with $f(x) < g(x)$ for every real number x, then $f'(x) < g'(x)$ for all x.

Then, which of the following is a correct statement?

(A) Both P(1) and P(2) are true. (B) Both P(1) and P(2) are false.
(C) P(1) is true and P(2) false. (D) P(1) is false and P(2) true.

Answer and Comments: (B). For P(1), take $f(x) = x$ and $g(x) = \sin(\frac{1}{x})$ for a counterexample. For P(2), take $f(x) = \sin x$ and $g(x) = 100$. ∎

P(1) would have been true if $\lim\limits_{x \to 0} f(x)$ were non-zero, because in that case $f(x) \neq 0$ in some sufficiently small deleted neighbourhood of 0 and we can write $g(x)$ as the ratio of the functions $f(x)g(x)$ and $f(x)$. The essential idea in dealing with P(2) is that a function can be made arbitrarily large by adding a constant to it, without increasing its derivative.

Q.22 The number of solutions of the equation $\sin 7x + \sin 3x = 0$ with $0 \leq x \leq 2\pi$ is

(A) 9 (B) 12 (C) 15 (D) 18.

Answer and Comments: (C). The given equation can be rewritten as

$$2\sin(5x)\cos(2x) = 0 \tag{1}$$

We find the zeros of $\sin 5x$ and of $\cos 2x$ separately.

If $\sin 5x = 0$, then $5x = k\pi$ for some integer k. Since $0 \leq x \leq 2\pi$, the possible values of k are 0 to 10. Each such value gives a solution and these 11 solutions are all distinct. The second equation $\cos 2x = 0$ has only four solutions in $[0, 2\pi]$, viz. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$ and $\frac{7\pi}{4}$. So, in all, (1) has $11 + 4$ i.e. 15 solutions. ∎

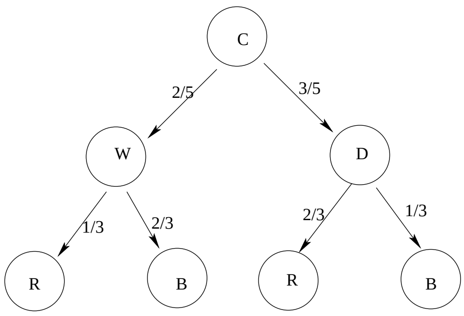
An absolutely simple problem. The last problem also involved solutions of $\sin x = 0$. But there is no comparison between their levels of difficulty. 14 should have been included as a fake answer to punish those who hastily conclude that $\sin 5x = 0$ has only 10 solutions.

Q.23 A bag contains some candies, $\frac{2}{5}$ of them are made of white chocolate and the remaining $\frac{3}{5}$ are made of dark chocolate. Out of the white chocolate candies, $1/3$ are wrapped in red paper, the rest are wrapped in blue paper. Out of the dark chocolate candies, $\frac{2}{3}$ are wrapped in red paper, the rest are wrapped in blue paper. If a randomly selected candy from the bag is found to be wrapped in red paper, then what is the probability that it is made up of dark chocolate?

(A) $\frac{2}{3}$ (B) $\frac{3}{4}$ (C) $\frac{3}{5}$ (D) $\frac{1}{4}$

Answer and Comments: (B). If the narrative in Q.3 (where Roger had 5 red tennis balls and 4 green tennis balls), was primary schoolish, the one in the present problem belongs to the kindergarten level!.

Anyway, coming to the problem itself, this is a problem about conditional probability. The following tree diagram summarises the data. The labels on the nodes and on the arrows are self explanatory.



Clearly,

$$\begin{aligned} P(R) &= \frac{2}{5} \times \frac{1}{3} + \frac{3}{5} \times \frac{2}{3} \\ &= \frac{2}{15} + \frac{6}{15} = \frac{8}{15} \end{aligned} \tag{1}$$

while

$$P(R \cap D) = \frac{3}{5} \times \frac{2}{3} = \frac{6}{15} \tag{2}$$

The problem asks for the conditional probability $P(D|R)$, i.e. the probability of D, given R. By Bayes theorem,

$$P(D|R) = \frac{P(R \cap D)}{P(R)} = \frac{6/15}{8/15} = \frac{3}{4} \tag{3}$$

So, this is the probability that a chocolate with a red wrapper happened to be dark. ∎

It takes more time to read the problem than to solve it! In fact, after getting (1), the calculations can be done mentally.

Such problems definitely do not enhance the prestige of a test in academic circles. Far better problems on conditional probability have been asked in JEE. See Comment No. 11 of Chapter 22, and especially the 1994 JEE problem there. In those days the candidates had to show their work. And sometimes it was a torture for the examiner to go through pages and pages of hastily scribbled, and often inconclusive, calculations with hardly any words of explanation. (Perhaps this was one of the reasons that led to the adoption of a totally objective type JEE). On this background, this particular problem was an examiner's dream. The correct answer to it was $\frac{193}{792}$. As soon as an examiner saw this magic figure displayed prominently at the end of a solution, he could safely skip the justification, because there is no way one can get this answer by fluke!

Q.24 A party is attended by twenty people. In any subset of four people, there is at least one person who knows the other three (we assume that if X knows Y, then Y knows X). Suppose there are three people in the party who do not know each other. How many people in the party know everyone?

(A) 16 (B) 17 (C) 18

(D) Cannot be determined from the given data.

Answer and Comments: (B). Forget about the use of the old fashioned 'people'. We shall call them as persons. Name them from P_1 to P_{20}. Suppose that P_1, P_2, P_3 are the ones who do not know each other. (It would be better to say, 'no two of whom know each other'.) Then, for every $i = 4, 5, \ldots, 20$, out of the four persons P_1, P_2, P_3 and P_i, someone knows all the other three from the first part of the data. This person has to be P_i since P_1, P_2, P_3 fail to know at least two persons in this foursome.

Let $S = \{P_4, P_5, \ldots, P_{20}\}$. What we have shown so far is that every $P_i \in S$ knows P_1, P_2, P_3. But we now claim that for any two (distinct) P_i and P_j in this set, P_i and P_j know each other. For otherwise in the foursome $\{P_1, P_2, P_i, P_j\}$, no one will know all the other three, contradicting the hypothesis.

Thus we have shown that all members of S know each other. Since they also know P_1, P_2, P_3, at least these 17 persons meet the condition of knowing everybody else. Since P_1, P_2, P_3 obviously fail to satisfy this condition, the answer is 17. ∎

A good problem, requiring simple deductive reasoning.

Q.25 The sum of all natural numbers a such that $a^2 - 16a + 67$ is a perfect square is:

(A) 10 (B) 12 (C) 16 (D) 22.

Answer and Comments: (C). Completing squares, the condition means

$$b^2 + 3 = c^2 \tag{1}$$

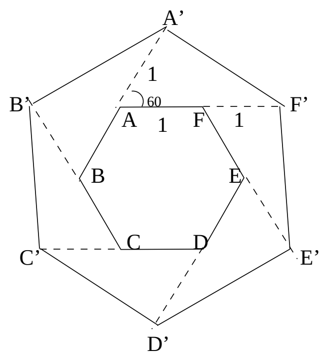
where $b = |a - 8|$ and c is some positive integer. This can be further rewritten as

$$(c - b)(c + b) = 3 \tag{2}$$

Since 3 is a prime, its only factorisation is 1×3. Equating the smaller factor 1 with $c - b$ and the larger one 3 with $c + b$ and solving the resulting system, we get $c = 2$ and $b = 1$. We are interested only in b. Having got $|a - 8| = 1$, a is either 7 or 9. So these are the only integral values of a which satisfy the given condition. Their sum is 16. ∎

Another problem which can be solved without much writing. And unlike the chocolate problem, here the time to read it is also negligible. So, although a trivial problem by itself, once in a while it is welcome, especially in an entrance test which is not solely based on such problems.

Q.26 The sides of a regular hexagon $ABCDEF$ are extended by doubling them (for example, BA extends to BA' with $BA' = 2BA$) to form a bigger regular hexagon $A'B'C'D'E'F'$ as in the figure.



Then the ratio of the areas of the bigger to the smaller hexagon is:

(A) 2 (B) 3 (C) $2\sqrt{3}$ (D) π.

Answer and Comments: (B). As both the hexagons are regular, the ratio of their areas will be the square of the ratio of their sides. Let us take the sides of the smaller hexagon 1 unit each. Then $AA' = 1$ and $AF' = AF + FF' = 2$. Also $\angle A'AF'$ equals $60°$, being an external angle of a regular hexagon. So, applying the cosine rule to the triangle $A'AF'$, we get

$$A'F'^2 = AA'^2 + AF'^2 - 2AA'.AF' \cos 60° = 5 - 2 = 3 \qquad (1)$$

So $A'F' = \sqrt{3}$. Therefore the ratio of the sides of the bigger to the smaller hexagons is $\sqrt{3}$. Hence the ratio of their areas is 3. ∎

A good, simple problem in geometry. By supplying the diagram, the paper-setters have ensured that it can be done within a reasonable time. A single application of the cosine rule is the only computation involved. So this problem is well suited for Paper 1.

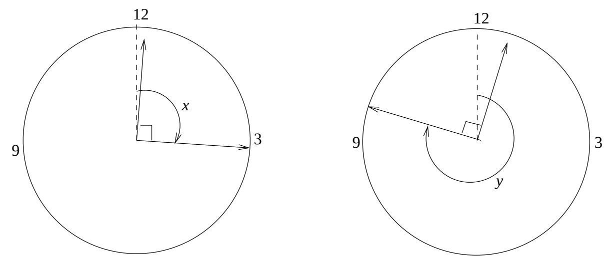
Q.27 Between 12 noon and 1 PM, there are two instants, when the hour hand and the minute hand of a clock are at right angles. The difference in minutes between these two instants is:

(A) $32\frac{8}{11}$ (B) $30\frac{8}{11}$ (C) $32\frac{5}{11}$ (D) $30\frac{5}{11}$.

Answer and Comments: (A). A typical high school problem. Let the two hands be at right angles for the first time x minutes after 12 noon. During this time the minute hand travels x minutes. But the hour hand travels only $\frac{x}{12}$ minutes. Being at right angles means an angular distance of 15 minutes. This gives an equation for x, viz.

$$x - \frac{x}{12} = 15 \tag{1}$$

solving which, we get $x = \frac{180}{11}$.



Now the two hands will again be at right angles after, say, y minutes past 12 noon. This time the minute hand has travelled y minutes and the hour hand only $\frac{y}{12}$ minutes. The angular distance between them is still 15 minutes. But now it is as if the minute hand is lagging behind the hour hand. So the equation for y is

$$y + 15 = 60 + \frac{y}{12} \tag{2}$$

Solving which we get $y = \frac{540}{11}$. Hence $y - x = \frac{360}{11} = 32\frac{8}{11}$. ∎

Another very simple problem. Its commendable feature is that it tests the ability to write down equations from a real life data. But the setting of a clock is very common in recreational puzzles in newspapers. A different setting based on, say, the orbital motion of celestial bodies would have been welcome.

Q.28 For which values of θ, with $0 < \theta < \pi/2$, does the quadratic polynomial in t given by $t^2 + 4t\cos\theta + \cot\theta$ have repeated roots?

(A) $\frac{\pi}{6}$ or $\frac{5\pi}{18}$ (B) $\frac{\pi}{6}$ or $\frac{5\pi}{12}$ (C) $\frac{\pi}{12}$ or $\frac{5\pi}{18}$ (D) $\frac{\pi}{12}$ or $\frac{5\pi}{12}$

Answer and Comments: (D). An ideal problem for those who hate word problems. You only have to know that a quadratic has repeated roots if and only if its discriminant vanishes. In the present problem, this means

$$16\cos^2\theta = 4\cot\theta \tag{1}$$

This resolves into two cases. (i) $\cos\theta = 0$ and (ii) $4\sin\theta\cos\theta = 1$. The first possibility gives nothing because $\cos\theta > 0$ for $0 < \theta < /pi/2$. (ii) simplifies to $\sin(2\theta) = \frac{1}{2}$. Since $2\theta \in (0, \pi)$, there are two possibilities, either $2\theta = \frac{\pi}{6}$ or $2\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$. Correspondingly, $\theta = \frac{\pi}{12}$ or $\frac{5\pi}{12}$. ∎

Another absolutely straightforward problem. The relationship with quadratic equations is superficial. Once it is removed, the problem vies with Q.22 for triviality. One really fails to know the purpose of such questions, both based on the same narrow area.

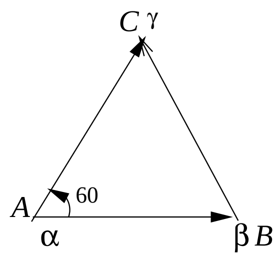
Q.29 Let α, β, γ be complex numbers which are the vertices of an equilateral triangle. Then we must have:

(A) $\alpha + \beta + \gamma = 0$ (B) $\alpha^2 + \beta^2 + \gamma^2 = 0$
(C) $\alpha^2 + \beta^2 + \gamma^2 + \alpha\beta + \beta\gamma + \gamma\alpha = 0$
(D) $(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2 = 0$

Answer and Comments: (D). Obviously, whatever be the right answer, it has to be cyclically symmetric in α, β, γ. As this is the case with all the four options, we cannot rule out any of them on this preliminary ground. But the first three options can be ruled out on a different ground. We can choose an equilateral triangle whose vertices are very close to each other but far away from 0. Then $\alpha+\beta+\gamma$ will be close to 3α but not 0. Similarly for $\alpha^2+\beta^2+\gamma^2$ and $\alpha^2+\beta^2+\gamma^2+\alpha\beta+\beta\gamma+\gamma\alpha$. Hence, by elimination, (D) is the correct option.

But instead of this back door entry, we give a proof that (D) holds if the complex numbers α, β, γ correspond to the vertices A, B, C

respectively of an equilateral triangle. Then the directed line segments $\overrightarrow{AC}$ and $\overrightarrow{AB}$ are represented, respectively, by the complex numbers $\gamma - \alpha$ and $\beta - \alpha$. But the former segment is obtained by rotating the latter around the point A through an angle $60°$. (In the diagram below this rotation is counterclockwise. The argument can be modified suitably, if it is clockwise.)



This observation gives us

$$\gamma - \alpha = e^{\pi i/3}(\beta - \alpha) \tag{1}$$

and hence by squaring,

$$(\gamma - \alpha)^2 = \omega(\alpha - \beta)^2 \tag{2}$$

where $\omega = e^{2\pi i/3} = \cos(\frac{2\pi}{3}) + i\sin(\frac{2\pi}{3})$.
By a similar reasoning (or by cyclical symmetry) and using (1)

$$(\beta - \gamma)^2 = \omega(\gamma - \alpha)^2 = \omega^2(\alpha - \beta)^2 \tag{3}$$

Adding,

$$(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2 = (1 + \omega + \omega^2)(\alpha - \beta)^2 = 0 \tag{4}$$

where the last step follows from the fact that $\omega^3 = 1$ but $\omega \neq 1$.

So, if α, β, γ represent the vertices of an equilateral triangle, then (D) holds. ∎

There are easier ways to derive (D). The simplest probably is to write $\overline{\alpha} - \overline{\beta}$ as $\dfrac{|\alpha - \beta|^2}{\alpha - \beta}$. Write $\overline{\beta} - \overline{\gamma}$ and $\overline{\gamma} - \overline{\alpha}$ similarly and add to get 0. As all numerators are equal (because of equilaterality), we get

$$\frac{1}{\alpha - \beta} + \frac{1}{\beta - \gamma} + \frac{1}{\gamma - \alpha} = 0 \tag{5}$$

which, on simplification gives

$$\alpha^2 + \beta^2 + \gamma^2 = \alpha\beta + \beta\gamma + \gamma\alpha \tag{6}$$

(D) follows by multiplying both the sides by 2 and rearranging. The converse is also true. That is, if (D) holds, then the triangle is equilateral. In fact, this characterisation of equilaterality is fairly well known (see Comment No. 12 of Chapter 8). Those who know it will get the answer instantaneously. To avoid the consequent inequity, this question should have been in Paper 2. If at all it is to be asked in the multiple choice format, care should have been taken to see that all the fake options cannot be eliminated without doing any work.

Q.30 Assume that n copies of the unit cube are glued together side by side to form a rectangular solid block. If the number of unit cubes that are completely invisible is 30, then the minimum possible value of n is:

(A) 204 (B) 180 (C) 140 (D) 84.

Answer and Comments: (C). Let the size of the rectangular block be $x \times y \times z$. Then we have

$$n = xyz \tag{1}$$

For a unit cube in this block to be completely invisible, it must not lie on any of the six surfaces. Put differently, it must lie in the 'core' of the block which is obtained by removing one layer of unit cubes on each surface. Since we are removing two layers of cubes parallel to each surface, the size of the core is $(x-2) \times (y-2) \times (z-2)$. We are given that

$$(x-2)(y-2)(z-2) = 30 \tag{2}$$

The problem asks to minimise (1) subject to the constraint (2). Here both the objective function (i.e. the function to be minimised) and the constraint are functions of three variables. If these variables can assume all possible positive real values, then we need calculus or some tricks such as the A.M.-G.M. inequality. But in the present problem, the dimensions of the block and hence of the core are integers. As 30 is a small number, it can be factored into a product of three positive integers in only a few ways. We can try all such factorisations, one by one, and calculate the value of n for each. For example the factorisation $6 \times 5 \times 1$ gives $x - 2 = 6, y - 2 = 5$ and $z - 2 = 1$. Hence $x = 8, y = 7, z = 3$. So $n = 168$. But intuitively, it is clear that to minimise n, the rectangular block should be as close to a cube as possible. 30 is not a perfect cube. But the factorisation $5 \times 3 \times 2$ is the closest we can come. For this factorisation, $n = 7 \times 5 \times 4 = 140$. ∎

A good problem, based on a realistic situation. Sometimes, we want to enclose some precious cubes made of gold, say, by a collection of relatively cheaper cubes of the same size, to protect against theft. The problem tells you how many cheaper cubes will be needed.

ISI BStat-BMath-UGA-2018 Paper 2 with Comments

There are EIGHT questions. Time allowed is 2 hours. Each question carries 10 marks.

Notations: In the following, $\mathbb{N} = \{1, 2, 3, \ldots\}$ denotes the set of all natural numbers, $\mathbb{R}$ denotes the set of real numbers.

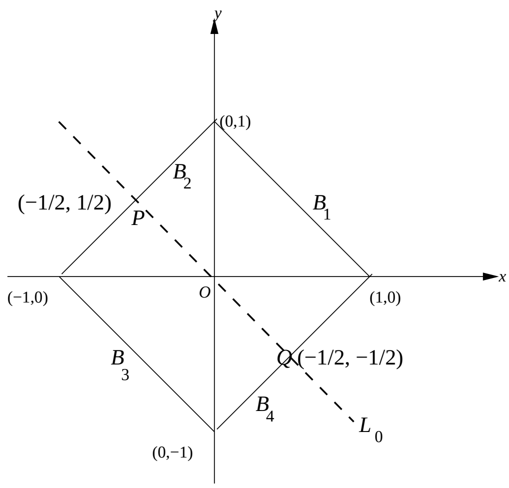
Q. 1 Find all pairs x, y with x, y real satisfying the equations:

$$\sin\left(\frac{x+y}{2}\right) = 0, \quad |x| + |y| = 1.$$

Solution and Comments: The best method is by drawing a diagram in which the solution set of each equation is shown. Let A be the solution set of the first equation. That is

$$A = \{(x, y) \in \mathbb{R}^2 : \sin\left(\frac{x+y}{2}\right) = 0\} \tag{1}$$

Then $(x, y) \in A$ if and only if $x + y = 2n\pi$ for some integer n. This is a disjoint union of an infinite family of parallel lines L_n for every integer n, where $L_n = \{(x, y) \in \mathbb{R}^2 : x + y = 2n\pi\}$. Only one such line, viz. $L_0 = \{(x, y) : x + y = 0\}$ is shown as a dotted thick line in the figure below.



The solution set, say B, of the second equation, viz. $|x| + |y| = 1$. consists of four line segments B_1, B_2, B_3 and B_4 depending upon the quadrant in which (x, y) lies. They are all shown in the figure.

It is cleat that L_0 intersects only B_2 and B_4 at the points $P = (-1/2, 1/2)$ and $Q = (1/2, -1/2)$. As for the other lines L_n, in the set A, they all lie outside the square B. (That is why they are not shown in the figure.) So, there are only two solutions to the system of equations, viz. $(-\frac{1}{2}, \frac{1}{2})$ and $(\frac{1}{2}, -\frac{1}{2})$. ∎

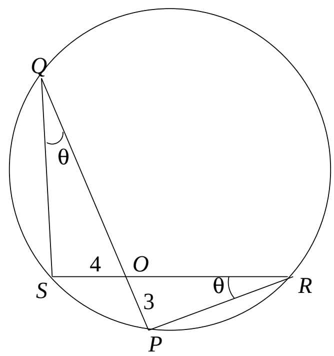
A fairly simple problem once the idea of drawing the solution sets of the two equations strikes.

Q. 2 Suppose PQ and RS are two chords of a circle intersecting at a point O. It is given that $PO = 3$ cm and $SO = 4$ cm. Moreover the area of the triangle POR is 7 cm^2. Find the area of the triangle QOS.

Solution and Comments: Since the quadrilateral $PRQS$ is cyclic,

$$\angle PRO = \angle SQO = \theta \text{ (say)} \tag{1}$$

Hence the triangles POR and SOQ are similar.



The ratio of the sides opposite to the angles θ is $\frac{3}{4}$. Hence the ratio of their areas is $\frac{9}{16}$. Therefore

$$\triangle QOS = \frac{16}{9}\Delta POR = \frac{16}{9} \times 7 = \frac{112}{9} \tag{2}$$

cm^2. ∎

A very simple problem. Should have been asked in Paper 1.

Q. 3 Let $f : I\!\!R \longrightarrow I\!\!R$ be a continuous function such that for all x and for all $t \geq 0$,

$$f(x) = f(e^t x).$$

Show that f is a constant function.

Solution and Comments: We first show that if x, y are of the same sign (i.e. both positive or both negative) then $f(x) = f(y)$. In this case both the ratios $\frac{y}{x}$ and $\frac{x}{y}$ are positive and one of them, say the first one, is at least 1. So taking $t = \log(\frac{y}{x})$ we have $y = e^t x$ and $t \geq 0$. Therefore from the hypothesis $f(x) = f(e^t x) = f(y)$.

Thus f is some constant, say C_1, on the positive x-axis and some constant C_2 on the negative x-axis. By continuity of f at 0, $C_1 = C_2$. But then f is constant on $I\!\!R$. ▮

Another very simple problem.

Q. 4 Let $f : (0, \infty) \longrightarrow I\!\!R$ be a continuous function such that for all $x \in (0, \infty)$,

$$f(2x) = f(x).$$

Show that the function g defined by the equation

$$g(x) = \int_x^{2x} f(t) \frac{dt}{t} \text{ for } x > 0$$

is a constant function.

Solution and Comments: The function $g(x)$ is a function defined by an integral. As the integrand is a continuous function of t and both the upper and lower limits of the integral (viz. $2x$ and x respectively), are differentiable functions of x, $g(x)$ is differentiable on $(0, \infty)$. Moreover, by Leibnitz rule (which is a combination of the Fundamental Theorem of Calculus and the Chain Rule), we have

$$\begin{aligned} g'(x) &= \frac{f(2x)}{2x}\frac{d}{dx}(2x) - \frac{f(x)}{x}\frac{d}{dx}(x) \\ &= \frac{f(2x)}{x} - \frac{f(x)}{x} \\ &= \frac{f(2x) - f(x)}{x} = 0 \end{aligned} \tag{1}$$

for all $x > 0$ from the hypothesis. So g' vanishes identically and hence g is constant. ▮

The proof is very similar to that of the assertion that if f is continuous and periodic with period T, then the integral $\int_a^{a+T} f(t)dt$ is independent of a, i.e. is a constant function of a. The result is still true even if f is not continuous, as long as it is integrable. But this proof becomes inapplicable.

Q. 5 Let $f : I\!R \longrightarrow I\!R$ be a differentiable function such that its derivative f' is a continuous function. Moreover, assume that for all $x \in I\!R$,

$$0 \leq |f'(x)| \leq \frac{1}{2}.$$

Define a sequence of real numbers $\{a_n\}_{n \in I\!N}$ by:

$$a_1 = 1$$

$$a_{n+1} = f(a_n) \text{ for all } n \in I\!N.$$

Prove that there exists a positive real number M such that for all $n \in I\!N$,

$$|a_n| \leq M.$$

Solution and Comments: In more common terminology, the problem asks to show that the sequence $\{a_n\}_{n \in I\!N}$ is bounded. In fact, the formulation would have been less clumsy had this term been used. It is avoided apparently to do justice to those candidates who may not know it. While such an intention, if at all present, is commendable, sometimes it has the very opposite effect. The concept of a bounded sequence is familiar to most candidates even at the HSC level and often used in statements such as 'Every convergent sequence is bounded, but not conversely'. When some commonly used term is replaced by its definition, one tends to get confused as to whether the paper setters have something else in mind. If you say 'the woman who gave birth to me', the first query would be 'You mean your mother?' and if you are not there to answer it, then the first suspicion is that although your mother gave birth to you, she probably died or abandoned you soon!

Let us leave aside this verbal objection and get down to the problem. We are given $a_1 = 1$ but not $a_2 = f(a_1)$ since the function f is not given. If $a_2 = a_1$, then $a_3 = f(a_2) = f(a_1)$ and by induction on n, it will follow that $a_n = a_1$ for all $n \in I\!N$. Then the sequence $\{a_n\}_{n \in I\!N}$ would be the constant sequence $\{1\}$ and we can take $M = 1$.

So we assume $a_2 = f(a_1) \neq a_1$. In that case, by Lagrange's Mean Value Theorem (LMVT), there exists some c_1 in the interval (a_1, a_2) (or the interval (a_2, a_1) as the case may be depending upon whether $a_1 < a_2$ ot $a_2 < a_1$), such that

$$a_3 - a_2 = f(a_2) - f(a_1) = f'(c_1)(a_2 - a_1) \tag{1}$$

Taking absolute values of both the sides and using the hypothesis about $|f'(x)|$, we get

$$|a_3 - a_2| = |f'(c_1)||a_2 - a_1| \leq \frac{1}{2}|a_2 - a_1| \tag{2}$$

We now repeat the argument for a_2 and a_3 to get some c_2 (in case $a_2 \neq a_3$) such that

$$\begin{aligned} |a_4 - a_3| &= |f'(c_2)||a_3 - a_2| \\ &\leq \frac{1}{2}|a_3 - a_2| \\ &\leq \frac{1}{4}|a_2 - a_1| \end{aligned} \tag{3}$$

where in the last step we have used (2).

By induction on n, it follows that

$$|a_n - a_{n-1}| \leq \frac{1}{2^{n-2}}|a_2 - a_1| \tag{4}$$

for all $n \geq 2$. (This includes the possibility that for some r, $a_{r+1} = f(a_r) = a_r$. If this happens then the L.H.S. of (4) is 0 for all $n \geq r+1$.)

As $|a_2 - a_1|$ is a fixed positive number, (4) shows that the consecutive terms of the sequence $\{a_n\}_{n \in I\!N}$ are getting closer and closer. This does not automatically prove that the sequence is bounded. A classic counterexample is the sequence of **harmonic numbers**, defined by,

$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$. Here $|H_n - H_{n-1}| = \frac{1}{n}$ which gets smaller and smaller as n increases. Still, it can be shown that the sequence $\{H_n\}_{n \in I\!N}$ is not bounded.

But the situation in our problem is more pleasant. Because of the factor 2^{n-2} in the denominator of the R.H.S. of (4), the difference between the consecutive terms of the sequence $\{a_n\}_{n \in I\!N}$ tends to 0 much more rapidly than $\frac{1}{n}$. In fact, using the triangle inequality and the formula for the sum of a geometric progression, we get

$$\begin{aligned}
|a_n - a_1| &= |(a_n - a_{n-1}) + (a_{n-1} - a_{n-2}) + \ldots + (a_3 - a_2) + (a_2 - a_1)| \\
&\leq \sum_{k=2}^{n} |a_k - a_{k-1}| \\
&\leq \sum_{k=2}^{n} \frac{1}{2^{k-2}} |a_2 - a_1| \qquad (5) \\
&= \frac{1 - \frac{1}{2^{n-1}}}{\frac{1}{2}} |a_2 - a_1| \qquad (6) \\
&\leq 2(|a_2 - a_1|) \qquad (7)
\end{aligned}$$

where in (5) we have used (4).

To finish the solution, we apply the triangle inequality again to get that for all $n \geq 2$,

$$|a_n| \leq |a_n - a_2| + |a_2| \leq 2|a_2 - a_1| + |a_2| \leq 3|a_2| + 2|a_1| \qquad (8)$$

As this also holds for $n = 1$, we can take M as $3|a_2| + 2|a_1|$. ∎

The essence of the solution was that by applying LMVT and the hypothesis we showed that for all $a, b \in I\!R$

$$|f(b) - f(a)| \leq \frac{1}{2}|b - a| \qquad (9)$$

and then to conclude that $\{a_n\}_{n \in I\!N}$ is bounded we crucially used that $\frac{1}{2} < 1$.

More generally, a function for which there exists some $\lambda \in [0, 1)$ such that for all $a, b \in I\!R$

$$|f(b) - f(a)| \leq \lambda|b - a| \qquad (10)$$

is called a **contraction**, because it brings the points of the domain closer. So, we have used LMVT and the hypothesis to show that f is a contraction. The subsequent part of the solution is applicable to any contraction. That is, if $f : \mathbb{R} \longrightarrow \mathbb{R}$ is a contraction, then starting with any $a_1 \in \mathbb{R}$, the recursively defined sequence $a_n = f(a_{n-1})$ for $n \geq 2$, is bounded. Actually, a stronger result is true. This sequence of iterates, as it is called, is not only bounded, but is convergent and its limit is the unique fixed point of f. This result is called the **contraction fixed point theorem**. The proof is also essentially the same, based on the fact that the geometric series $\sum\limits_{n=0}^{\infty} \lambda^n$ is convergent for $|\lambda| < 1$.

The contraction fixed point theorem also holds for functions $\mathbf{f}$ from $\mathbb{R}^n$ to $\mathbb{R}^n$. The only change is that in the the definition of contraction, (1) is replaced by

$$\|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\| \leq \lambda \|\mathbf{b} - \mathbf{a}\| \tag{11}$$

for all $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. That is, we replace the absolute value by the euclidean distance. In fact, the theorem holds more generally, for what are called **complete metric spaces** and has important applications in proving the existencxe of solutions of certain differential equations.

We note in passing that the contraction fixed point theorem may fail if (10) is replaced by the weaker hypothesis

$$|f(b) - f(a)| < |b - a| \tag{12}$$

for all a, b in the domain. For example, the function $f : [2, \infty) \longrightarrow [2, \infty)$ defined by $f(x) = x + \frac{1}{x}$ satisfies (12) as we see by applying LMVT and the fact that $f'(x) = 1 - \frac{1}{x^2} < 1$ for all $x \in [0, \infty)$. A function which satisfies (12) is called a **weak contraction**.

Q. 6 Let $a \geq b \geq c > 0$ be real numbers such that for all $n \in \mathbb{N}$, there exist triangles of side lengths a^n, b^n, c^n. Prove that the triangles are isosceles.

Solution and Comments: It suffices to show that $a = b$. For every $n \in \mathbb{N}$, we have $a^n \geq b^n \geq c^n > 0$. They will form a triangle if and only if $b^n + c^n > a^n$ which translates as

$$u^n + v^n > 1 \tag{1}$$

where $u = \frac{b}{a}$ and $v = \frac{c}{a}$.

Now, if $a > b$ then $a > c$ and so both u, v will lie in the interval $(0, 1)$. But then u^n as well as v^n tends to 0 as $n \to \infty$. That will be a contradition if we take limits of both the sides of (1) as $n \to \infty$. So $a = b$. Therefore for every n, the triangle with sides a^n, b^n, c^n is isosceles. ∎

A simple problem based on an interesting application of the fact that $x^n \to 0$ if $|x| < 1$. Note that we cannot assert that the triangles are equilateral. In fact, for any $a > c > 0$, we can form a triangle with sides a^n, a^n and c^n.

Q. 7 Let $a, b, c \in I\!N$ be such that

$$a^2 + b^2 = c^2 \text{ and } c - b = 1.$$

Prove that

(i) a is odd,

(ii) b is divisible by 4,

(iii) $a^b + b^a$ is divisible by c.

Solution and Comments: Since $c = b + 1$, a direct substitution into the equation given gives

$$a^2 = 2b + 1 \tag{1}$$

which shows that a^2 and hence a is odd. Also from (1) we get

$$2b = (a - 1)(a + 1) \tag{2}$$

The integers $a - 1$ and $a + 1$ is a pair of consecutive even integers. So one of them is divisible by 4. Hence $2b$ is divisible by 8, whence b is divisible by 4.

It only remains to prove (iii). Since $b = c - 1$, we can rewrite (1) as

$$a^2 = 2c - 1 \tag{3}$$

We already showed that $b = 4m$ for some positive integer m. So, from (3),

$$a^b = a^{4m} = (a^2)^{2m} = (2c-1)^{2m} \tag{4}$$

When the last term is expanded by the binomial theorem, all terms except the constant term will be divisible by c (in fact by c^2). The constant term is $(-1)^{2m}$ which is simply 1. So we have

$$a^b = pc + 1 \tag{5}$$

for some integer p. As for b^a, we write b as $c-1$. Then

$$b^a = (c-1)^a \tag{6}$$

once again all the terms in the binomial expansion of the R.H.S., except the constant term are divisible by c. But this time the constant term is $(-1)^a$ which equals -1 since a is odd. Therefore

$$b^a = qc - 1 \tag{7}$$

for some integer q.

From (5) and (7),

$$a^b + b^a = (p+q)c \tag{8}$$

which proves (iii). ∎

A simple problem about Pythagorean triplets. Examples of such triples are $(3, 4, 5)$ and $(5, 12, 13)$. But because of the restriction that $c - b = 1$, the general form of the Pythagorean triplets given in the comments to Q.17 of Paper 1 is not needed here and that makes the problem elementary. A more sophisticated formulation of the solution to (iii) is possible if we use the concept of congruency modulo c. In essence, the proof of (iii) boils down to showing that $a^b \equiv 1$ *modulo* c and $b^a \equiv -1$ *modulo* c. But we have avoided this terminology and worked directly using the binomial theorem.

Q. 8 Let $n \geq 3$. Let $A = ((a_{ij}))_{1 \leq i,j \leq n}$ be an $n \times n$ matrix such that $a_{ij} \in \{1, -1\}$ for all $1 \leq i, j \leq n$. Suppose that

$$a_{k1} = 1 \text{ for all } 1 \leq k \leq n \text{ and}$$

$$\sum_{k=1}^{n} a_{ki}a_{kj} = 0 \text{ for all } i \neq j.$$

Show that n is a multiple of 4.

Solution and Comments: We are given that the first column of A has all 1's in it. The second part of the hypothesis is popularly expressed by saying that every two distinct columns of A are **orthogonal** to each other, because their inner product (regarded as n-dimensional column vectors) is 0. Since every column of A other than the first column is orthogonal to the first column and has only 1 and -1 as possible entries, exactly half of them are 1 and the other half must be -1. In particular, this shows that n is even, say $n = 2m$. Our task is to show that m is itself even.

For this we normalise the second column of the matrix A. We know that exactly m entries of the second column of A are 1 and the remaining are -1. Since any reshuffling of rows does not affect the orthogonality condition, we may assume, without loss of generality that the first m entries of the second column are 1 and the last m entries are -1. Thus, if $n = 4$, the submatrix of A consisting of the first two columns is

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \\ 1 & -1 \end{bmatrix} \tag{1}$$

Now consider the third column of A, which exists because it is given that $n \geq 3$. Because of orthogonality with the first column, m of its entries are 1 and the remaining m entries are -1. This we already know. Let r be the number of the first m entries in the third column which equal 1. Then the remaining $m-r$ entries among the first entries equal -1 each. Also in the last m entries of the third column, there must be $m-r$ entries that equal 1 and r entries that equal -1 (because the total of each type is m). We already know the entries of the (normalised) second column. But now the inner product of the third column and the second column can be split into an 'upper' and a 'lower' part as

$$\sum_{k=1}^{n} a_{k2}a_{k3} = \sum_{k=1}^{2m} a_{k2}a_{k3} = \sum_{k=1}^{m} a_{k2}a_{k3} + \sum_{k=m+1}^{2m} a_{k2}a_{k3} \tag{2}$$

In the first summation in the last term, $a_{k2} = 1$ for all $1 \leq k \leq m$. But a_{k3} equals 1 for r values of k and equals -1 for the remaining $m - r$ values of k. We don't know exactly which ones are 1 and which ones are -1. But regardless of that, the 'upper' part of the inner product equals

$$\sum_{k=1}^{m} a_{k2}a_{k3} = r - (m - r) = 2r - m \tag{3}$$

On the other hand, for the second summation of the last term in (2), $a_{k2} = -1$ for all $m+1 \leq k \leq m$. So by an analogous reasoning we have

$$\sum_{k=m+1}^{2m} a_{k2}a_{k3} = r - (m - r) = 2r - m \tag{4}$$

As the leftmost term of (2) is given to be 0, we get

$$0 = 2r - m + 2r - m = 4r - 2m \tag{5}$$

which shows that $m = 2r$ and hence $n = 4r$. Thus n is divisible by 4. ∎

We can complete (1) to a 4×4 matrix A

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \tag{6}$$

which satisfies all the conditions in the data.

Matrices which satisfy the conditions in the problem are called **Hadamard matrices**. We just gave an example, of a 'smallest' Hadamard matrix. There is a way to build larger Hadamard matrices starting from smaller ones, by taking what is called their **tensor product**. So Hadamard matrices of arbitrarily large size are possible to construct. Interested readers will find the details on p.704 of the author's *Applied Discrete Structures*, New Age International (P) Ltd., New Delhi.

CONCLUDING REMARKS

By and large, most problems this year are simpler as compared with those in the ISI entrance tests of recent years. In fact, many of them are trivial. This includes Q.2 (about teams of officers), Q.3 (about Roger and the tennis balls), Q.12 (about symmetric arrangement of balls), Q.23 (about conditional probability) and Q.25 (about a quadratic assuming integral values). Both the problems (Q.22 and Q.28) on trigonometric equations are absolutely straightforward.

A few questions in Paper 1 are trivial, once the essential idea is correctly grasped. The masterpiece is Q.8 where the condition for an arithmetic progression is given in an intentionally clumsy form. Q. 13 and Q.14 also falls under this category. Q.15 has an ambiguous formulation. Once it is understood, it is a high school problem about time, work and rate. The non-existence of any solutions to the diophantine equation in Q.17 also hinges simply on parity considerations.

As it happens every year, there are a few questions in Paper 1, where the correct answer can be identified simply by eliminating the wrong options. These include Q.5 (about the possible numbers of roots and local extrema of a fourth degree polynomial). Q.7 (about matrices satisfying $A^4 = A$) and, most prominently, Q.29 (about characterisation of equilateral triangles in terms of complex numbers).

Among the better questions in Paper 1, Q.18 is probably the best. It combines the representation of an integer in a given base and logarithms. Q.10 is also good but has an unintended easy solution using power series for $\sin x$ and $\cos x$. Q.4 (about permutations which move every point by at most 1), Q.16 (about maximising the area of a rectangle whose sides contain the vertices of a given rectangle), Q.19 (about finiteness of the number of solutions of $\sin x/x = c$) and Q.27 (about the two hands of a clock being at right angles) are more suitable in Paper 2, with the modifications suggested. Q.24 gives the impression that graph theory will be useful. But the solution requires only pure deductive logic. Q.30 is a nice problem of constrained integral optimisation in a real life situation.

Over-all, Paper 1 is a mixed bag. Once again one wishes that by removing some absolutely easy questions and retaining only 15 to 20 relatively good ones, the test would have been a more valid measure of a candidate's abilities.

It is Paper 2 that is really disappointing in comparison with the past years. There is not even one question which can be called unusual and really makes you think. The first four questions belong more to Paper 1. The

Pythagorean triplets in Q.7 are too restricted and do not need their general structure. Instead, the problem reduces to a divisibility problem. Q.5 and Q.8 are forerunners of the contraction fixed point theorem and Hadamard matrices respectively. The former is based on the fact that $x^n \to 0$ as $x \to \infty$ if $|x| < 1$. Q.6 is also a cute application of this fact. This duplication could have been avoided. Instead, a question about inequalities would have been welcome as this topic is not even paid a lip service this year.